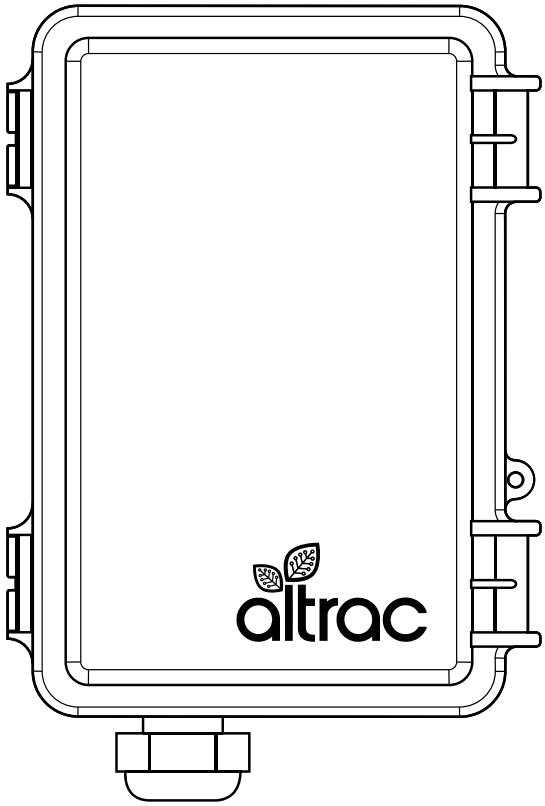
Wind Machine Station



Model Number:

ST100_3G

ST100_5G

User's Manual

Revised Jan 20, 2021

20885 Redwood Road, Box 218
Castro Valley, CA 94546
(510) 248-4141 | altrac.io

Table of Contents

Description and Specifications..... 3
Mounting Options..... 10
Sensors 13
- Temperature..... 13
- Fuel Level..... 15

Safety Information..... 17
Installation Instructions..... 18
- Orchard Rite 360, 454, 460..... 18
- Orchard Rite Ford V-10..... 26
- Orchard Rite Ford 6.2L V-8..... 34
- Orchard Rite CAT 4.4, 6,6, 7.1..... 42
- Amarillo Ford V-10/V-8..... 50
- Chinook/Hauff Ford V-10..... 57
- Vamco 360, 454, 460..... 65
- Electric..... 72

Software..... 78
- Webapp Operating Instructions 78
- RPM Calibration..... 79
- Fuel Sensor Calibration..... 80
- Installation Testing and Finaling..... 81
- Device and Tile States..... 82

Hardware..... 92
- Connectivity..... 92
- Battery..... 97
- Warranty 98

Please contact us if you have any questions.
Web: https://altrac.io/support
Phone: (510) 248-4141
Email: support@altrac.io

Consumer Safety Information



This is the safety alert symbol. This symbol alerts you to potential hazards that can kill or hurt you or damage the ST100 Wind Machine Station or associated equipment.

Safety Behavior and Practices

• Improper installation could result in shock or fire hazard. If you are uncomfotable with the installation contact support@altrac.io.

• Do not use this device if any part has been under water. Immediately call a qualified service technician to inspect the device and to replace any part of the control system that is damaged.

• Any alteration to the device or its controls can be dangerous and will void the warranty.

Safety and Reliability Features

• During installation or maintenance, you should unplug the ST100 from the engine wiring harness to ensure the ST100 does not turn on the wind machine automatically.

• The ST100 stores start and stop temperature data on the device. In case of a cellular failure, the ST100 will continue running according to the latest downloaded settings until it reconnects and downloads new settings.

• The ST100 utilizes automotive grade components to resist vibration and heat damage.

ST100 Key Components



① **ST100 Enclosure**

② **External Antenna Coax**

③ **Control Line**

④ **12-Pin Wiring Harness Connector**

⑤ **2-pin Temperature Probe Connector**

Specifications

Model	ST100_3G	ST100_5G
Cellular Modem	2G/3G	LTE CATM1 (5G)
Cellular Networks	AT&T & T-mobile	AT&T
Power Supply	12 VDC	
Battery	3.7V, 1800mAh	
Peak Electrical Consumption	50mA @ 12VDC	
Wiring	20/12, Direct Burial Rated, UV Stabilized	
Operating Temperatures	-40°C to 80°C (-40°F to 176°F)	
Weatherproofing	IP66, NEMA 1,2,3,3R,4,4X,12,13	
Enclosure Material	PBT/PC Blended Plastic	
UL Rating	UL94V-0 Flammability Rating	
Sunlight Protection	UV Stabilized	

Altrac is continually updating and improving products; specifications are subject to change without prior notice.

Description of Operation

The wind machine station has three components, (a) the cellular powered station which attaches to the wind machine tower to send and receive data (b) the engine wiring harness which plugs into your wind machine engine and autostart to gather sensor data (c) a temperature probe that reads ambient air temperature. The station can be installed on any wind machine make or model although the wiring harness will differ between machines.

Other optional sensors exist such as soil moisture and fuel level. If needed, Altrac can provide an external antenna to increase cellular signal if current coverage is not sufficent.

All the information gathered by the Altrac station is accessed through the Altrac webapp. From the webapp you can monitor engine status, turn the engine on/off, and set up alerts and management reports. Visit altrac.io for more information.

Features

Feature	Standard Software
Connection Frequency	Every 15 minutes when wind machine is OFF. Every 60-90 second when wind machine is ON or ambient air temperature is within 2 degrees of stop temperature.
Webapp Access	Included
Ambient Air Temperature	Included
Engine RPM	Included
Battery Voltage	Included
Hand-Off-Auto Switch Position	Included
Fuel Level	Diesel or LPG (Propane) Sensor Required
Soil Moisture	Soil Moisture Sensor Required
Alerts: Text, Email, Phone	Included
Management Reports: Daily, Yearly	Included

Standard Installation

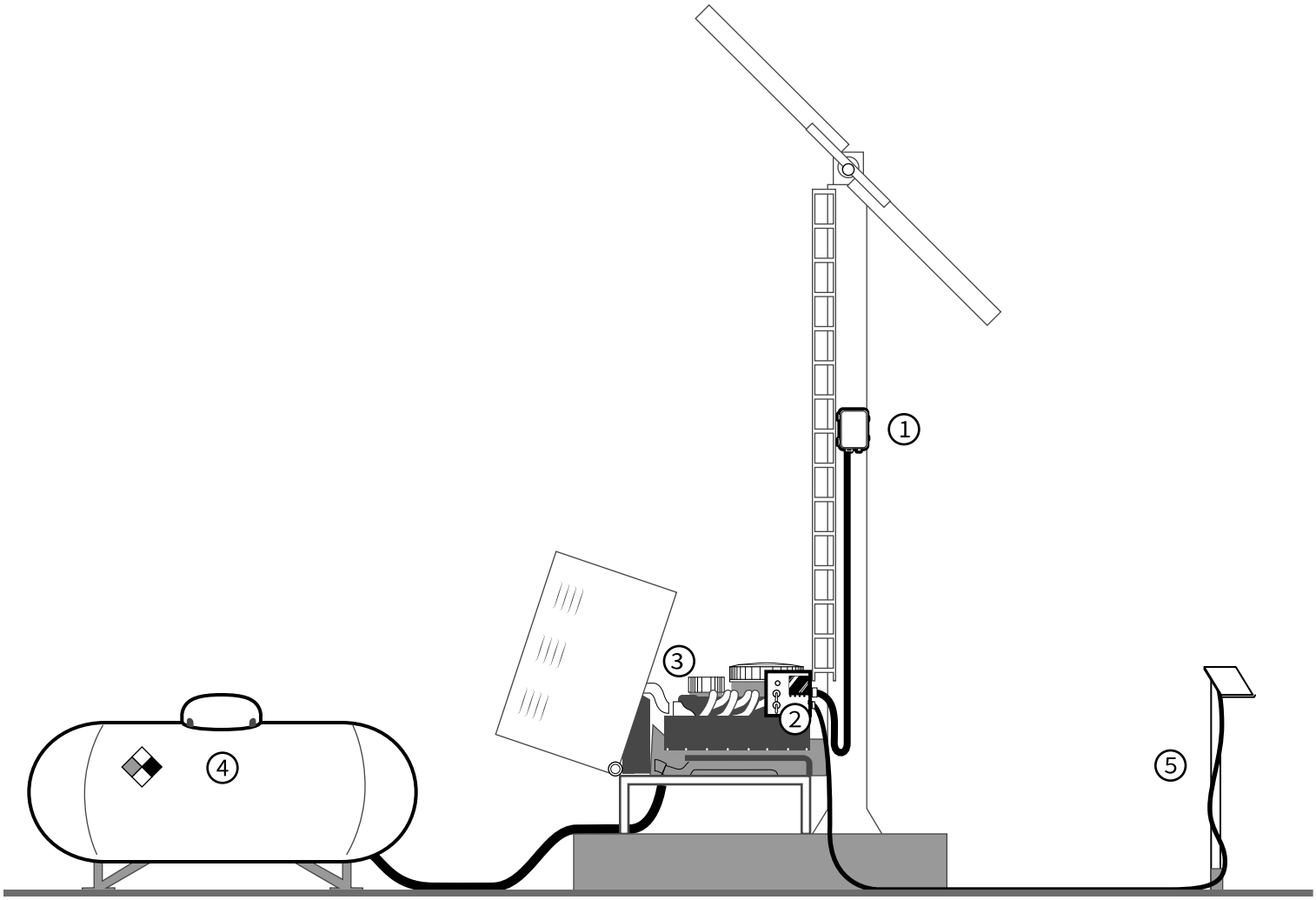
① **ST100**
Cellular-powered station attaches to the wind machine tower or a pole near the machine. This station records, sends, and recieves sensor data.

② **Engine**
The engine powers the wind machine fan.

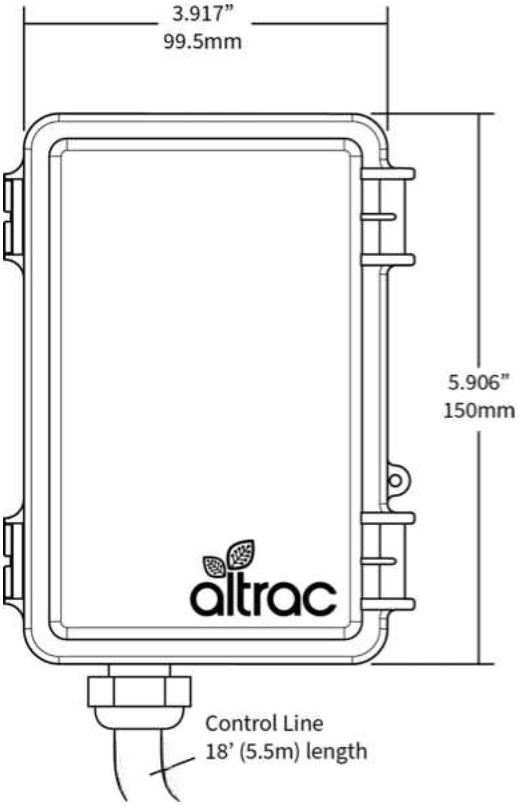
③ **Engine Wiring Harness**
The engine wiring harness attaches to various points on the engine to collect sensor data. Each wind machine type has a unique harness.

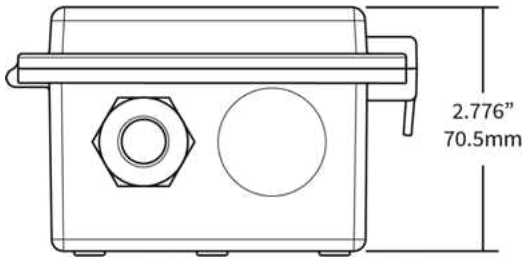
④ **Fuel Sensor**
The propane fuel sensor attaches to the gauge on the LPG tank. For diesel powered machines, the sensor is lowered into the bottom of the tank.

⑤ **Temperature Probe**
The Altrac temperature probe takes readings of the ambient air temperature. This probe should be protected from direct sunlight.



Dimensions and Components





Bracket Specifications

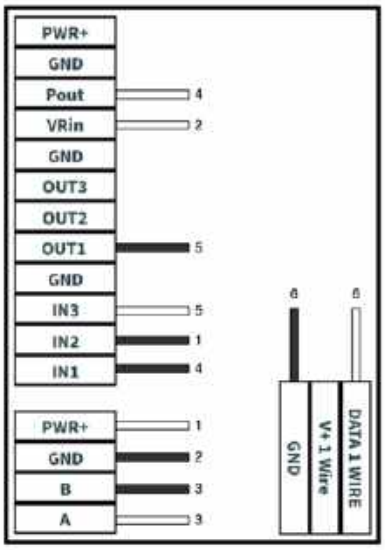
Brackets are designed to attach to (a) the ladder on a wind machine or (b) a 1.5" to 2.0" pipe.



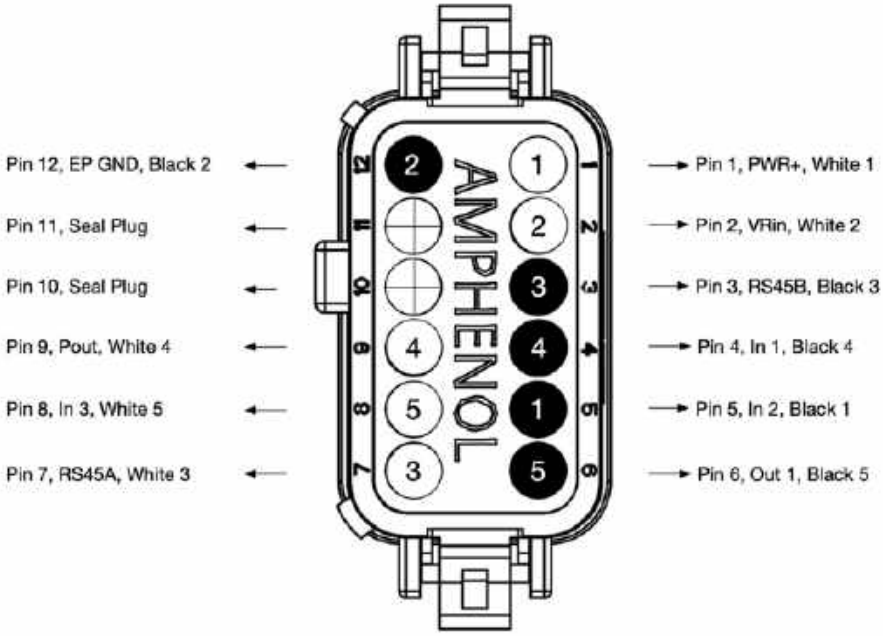
Wiring Schematic

EP I/O	Connector	Connector Pin #	Conductor Pair #	Conductor Color
RS-485 A	Amphenol 12P	7	3	White
RS-485 B	Amphenol 12P	3	3	Black
GND	Amphenol 12P	12	2	Black
PWR +	Amphenol 12P	1	1	White
IN 1	Amphenol 12P	4	4	Black
IN 2	Amphenol 12P	5	1	Black
IN 3	Amphenol 12P	8	5	White
OUT 1	Amphenol 12P	6	5	Black
VR IN	Amphenol 12P	2	2	White
P OUT	Amphenol 12P	9	4	White
GND (Temp)	Furukawa 2P	2	6	Black
DATA 1 WIRE	Furukawa 2P	1	6	White

EP3.5
PCB Wiring: Universal
Control Line: WH200_2.1
Enclosure: 1.2B

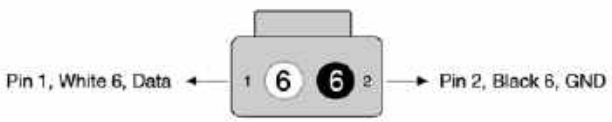


A
12 PIN Amphenol Connector
Female Pins



View from Back

B
2 Pin Furukawa Connector
Female Pins



View from Back

Mounting Installation

The Altrac station comes with pre-assembled mounting hardware. The ST100 is designed to be mounted on your wind machine tower with the contol line running down the interior of the tower. Alternatively, the ST100 can be mounted to a pole near the wind machine.





Mounting Location

- As high above the ground as practical.
- In an area clear of vegetation.

① Notice that each mounting hardware set has the hex bolt facing the same way but the brackets are mirrored.



② Using a phillips screw driver screw on the brackets to the bud box using the preexisting holes found on the back of the bud box.



③ Finished assembly should look like the picture to the right.



⑥ Fasten hose clamps to the vertical bar of the wind machine ladder.



⑦ Loop control line over driveshaft cross bar and zip tie close to the inner wall of the wind machine tower. Keep the cable away from the rotating shaft of the wind machine drive line.

⑧ Run the control line down the interior of the tower. Pay attention that the cable is not wound around the driveshaft. At the bottom of the tower, secure the cable to the gear box or drill a small hole in the tower wall and ziptie to secure.



Altrac Temperature Probe (SN100)



① **Temperature Probe**

② **2-Pin Connector attaches to Altrac Station**

	Specification
Length	50' (15m)
Max Length (with Extension)	150' (45m)
Min Temp	-67*F (-55*C)
Max Temp	257*F (125*C)
Accuracy	±1*F (±.5*C)
Sheath	Nylon sheath for rodent protection.
Calibration	Pre-calibrated. No need to calibrate.

Extension Cables (SN100_EXT_X)

Altrac offers extension cables for the SN100. Each extension comes with two connectors in order to create an easy, waterproof connection. Contact sales@altrac.io for more information.

Temperature Probe Installation

The SN100 temperature probe uses a digital signal to communicate with the Altrac device and must be plugged into the Altrac station to function.

STEP 1
Place the tip of the temperature probe under the sun shield and secure with zipties, etc.



STEP 2
Plug the black 2-pin connector into the mating connector on the ST100 Control Line.



Altrac LPG-Propane Fuel Sensor (SN110)



① **Hall Effect Sensor (Plugs into dial on tank)**

② **6-Pin Deutsch Connector (plugs into Accessory Port on Altrac wiring harness)**

③ **Intrinsically Safe Barrier Device**

	Specification
Length	40' (12m)
Max Length (with Extension)	500' (150m)
Min Propane Level	Inaccurate below 20%
Max Propane Level	Inaccurate above 80%
Accuracy	±5%
Sheath	Nylon sheath for rodent protection.
Calibration	Pre-calibrated. No need to calibrate.

Extension Cables (SN110_EXT_X)

Altrac offers extension cables for the SN110. Each extension comes with two connectors in order to create an easy, waterproof connection. Contact sales@altrac.io for more information.

Installation Instructions

Visit altrac.io/support to view installation instructions or email support@altrac.io

Altrac Diesel Fuel Sensor (SN111)



① **Junction Box with Vent**

② **Liquid Level Sensor**

③ **6-Pin Deutsch Connector (Plugs into Accessory Port)**

	Specifications
Length	40' (12m)
Max Length (with Extension)	500' (150m)
Measurement Medium	Water or Diesel Fuel
Min Liquid Level	Inaccurate below 10%
Max Liquid Level	6.5' (2m)
Accuracy	±.5%
Working Temperature Range	-30 to 80*C
Signal Output	.5 - 4.5VDC
Cable Type	PUR Cable (Oil Compatible)
Protection	IP68
Calibration	Pre-calibrated. No need to calibrate

Extension Cables (SN110_EXT_X)

Altrac offers extension cables for the SN110. Each extension comes with two connectors in order to create an easy, waterproof connection. Contact sales@altrac.io for more information.

Installation Instructions

Visit altrac.io/support to view installation instructions or email support@altrac.io

Wind Machine Safety Information

Before you attempt a wiring harness installation, you must follow the instructions below. Failure to follow the instructions will void the warranty on your device and wiring harness.

① Turn wind machine engine OFF by setting the Manual-Off-Auto switch to the 'OFF' position. Never install or perform maintenance on the wind machine while it is running.



② Disconnect the wind machine 12V battery.



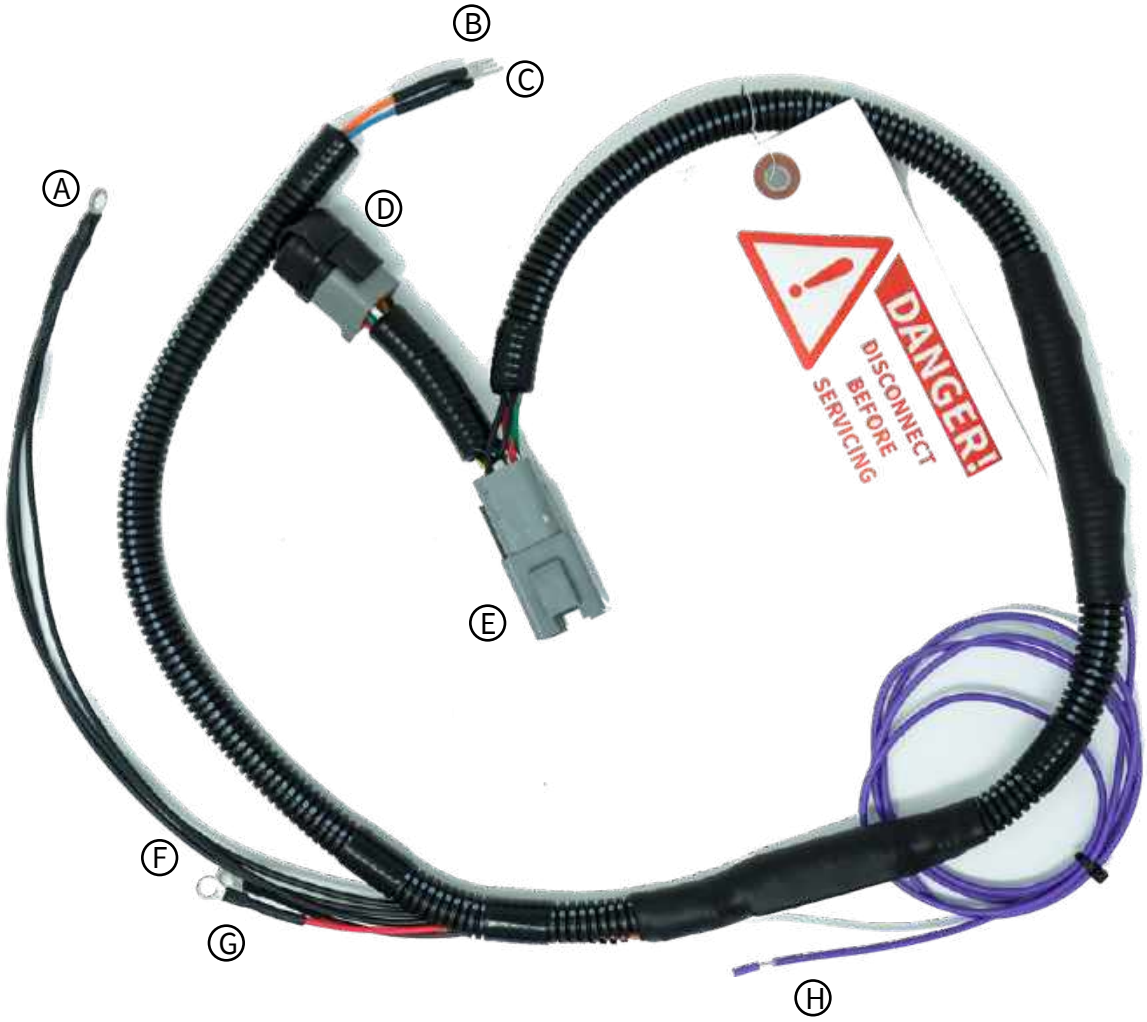
Make: Ochard Rite
Engine: 360, 454, 460
Autostart: Orchard Rite
Altrac Wiring Harness: WH102

Learn how to install the Altrac wiring harness for the engine below. Once installed, the Altrac Station will have access to power as well as readings from various sensors on the engine.



Before you attempt a wiring harness installation, you must unhook the wind machine battery and turn the Manual-Off-Auto switch to the OFF position.

WH102



Key	Wire Color	Purpose	Terminal Type	Installation
A	Black	Ground	#10 Ring	Attach to wind machine ground.
B	Orange	Power for Start Signal	#8 Fork	Replaces Orchard Rite temperature probe's red wire.
C	Blue	Start Signal	#8 Fork	Replaces Orchard Rite temperature probe's black wire.
D	Multiple		6-Pin Deutsch Connector	Plug in fuel sensors, soil moisture sensors, etc.
E	Multiple		12-Pin Deutsch Connector	Plug into the matching 12-pin connector on the ST100's control line.
F	Red	12VDC Power (Battery Voltage)	#10 Ring Terminal	Connect to bottom-middle terminal on Manual-Off-Auto switch.
G	Brown	Auto Switch Detection (12VDC)	#10 Ring Terminal	Connect to the bottom-left ('Auto' side) of the Manual-Off-Auto switch.
H	Purple	Tach Sensor (RPM)	18AWG GXL Wire	Splice into the tach signal from the distributor cap.

Tool List

Use	Tool
Wiring Harness Installation	Philips Screwdriver - Medium
Wiring Harness Installation	Philips Screwdriver - Large
Wiring Harness Installation	Flathead Screwdriver - Large
Wiring Harness Installation	Magnetic Bolt Retriever
Wiring Harness Installation	Channel Locks
Wiring Harness Installation	Electrical Tape
Wiring Harness Installation	4" Zip Ties
Wiring Harness Installation	12" Zip Ties
Wiring Harness Installation	18" Zip Ties
Wiring Harness Installation	Nippers (Flush Cutters)
Wiring Harness Installation	Adjustable Cresent Wrench
Wiring Harness Installation	Socket Drivers
Wiring Harness Installation	Wire Strippers
Wiring Harness Installation	Crimpers for Butt Connector
Wiring Harness Installation	Butt Connector (Blue)
Device Mounting	Impact Driver/Drill
Device Mounting	Metric Socket Set for Drill
Device Mounting	Drill Bits and Drivers
Device Mounting	Ladder
Device Mounting	Body Harness

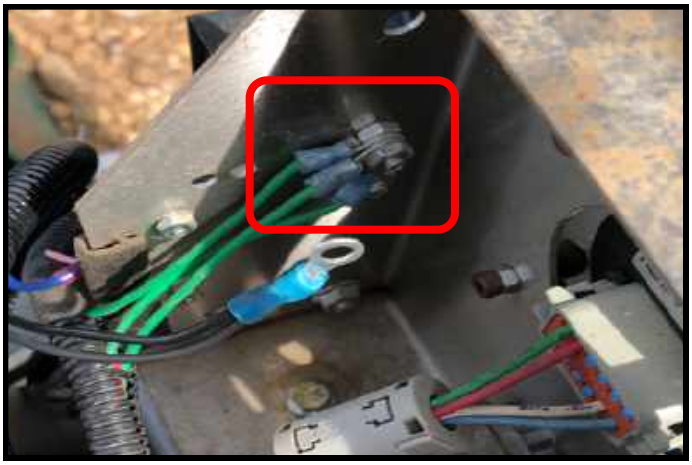


Before you attempt a wiring harness installation, you must unhook the wind machine battery and turn the Manual-Off-Auto switch to the OFF position.

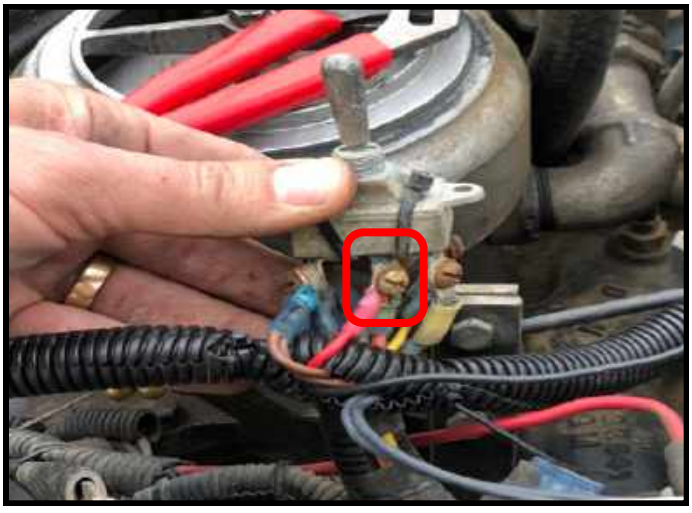
① Disconnect the wind machine engine's 12V battery.



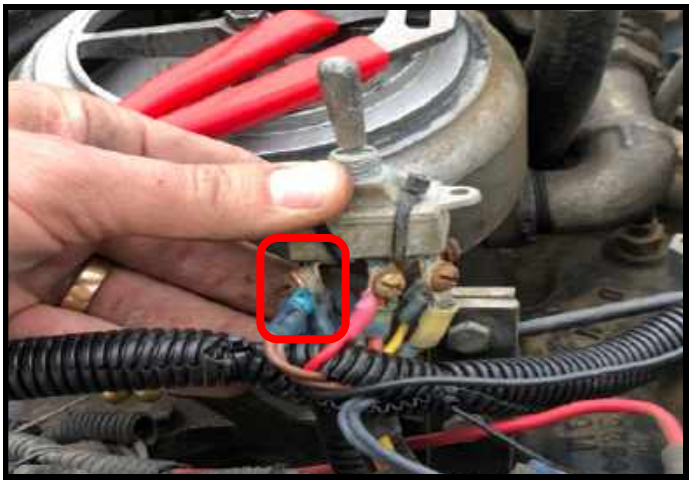
② Connect black ground wire to a ground source on engine. Other ground wires are usually green or black.



③ Connect Red Wire to the bottom-middle contact of the MAN-OFF-AUTO switch on the wind machine gauge panel. You will need to remove the switch with a crescent wrench.

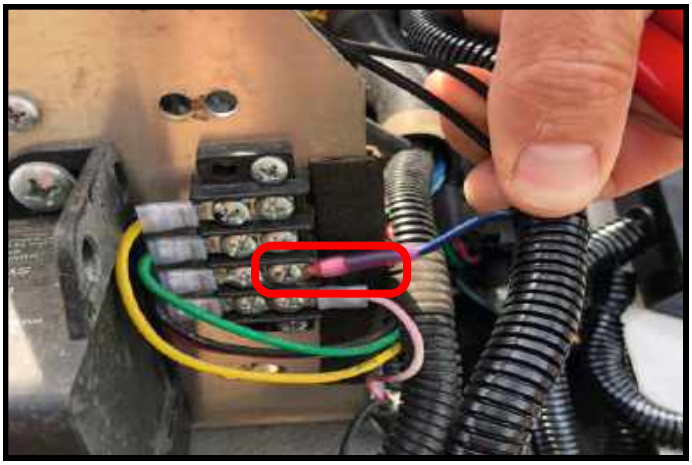


④ Connect Brown Wire to the bottom-left contact of the MAN-OFF-AUTO switch on the wind machine gauge panel. You will need to remove the switch with a crescent wrench.



⑤ Replace the black wire from Orchard Rite temperature probe with blue wire from harness.

HINT: Follow the temperature probe wire to the terminal block on the gauge panel to find the correct wires.



⑥ Replace red wire from Orchard Rite temperature probe with orange wire from harness.

HINT: Follow the temperature probe wire to the terminal block on the gauge panel to find the correct wires.



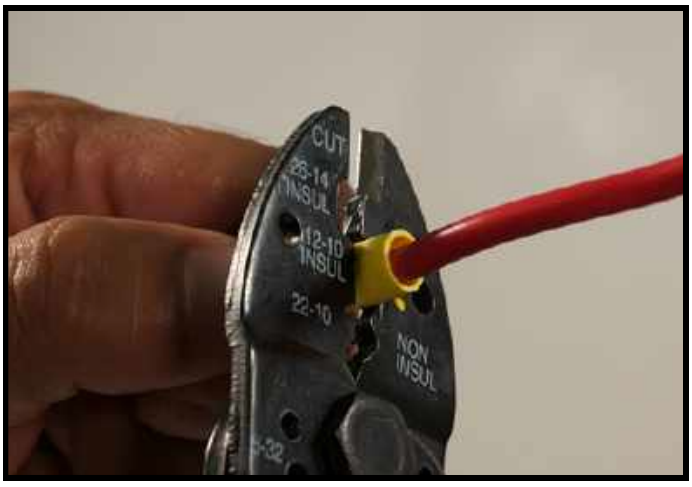
⑦ Splice the purple wire into the tach output on your coil. The wire is usally white and labeled TACH.



HINT: There are many places the autostart picks up a tach signal. Find the small black box on the gauge panel and look for a white wire. This white wire is always the RPM signal.



⑧ Cut the the white wire and splice in the purple Tach wire. Two wires will go in one side and one in the other.



⑨ Reattach the wind machine battery.

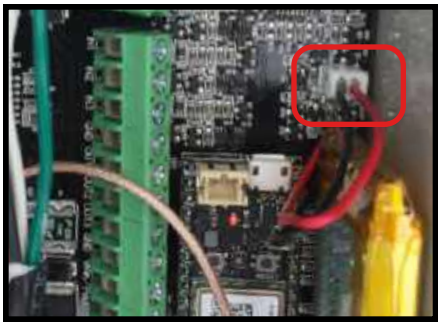


⑩ Connect the new wiring harness to the Altrac Station via the 12-pin connector on the control line.

HINT: You should tuck the connector up under the wind machine hood when you are finished installing to maximize weather protection.



① Plug in LiPo battery in ST100. The small silver battery in the enclosure connects directly to the black circuit board.



⑨ To run the wind machine you must have the Manual-Off-Auto switch in the 'Auto' position, and the autostart must be armed. You will see a blinking white light when the autostart is armed.



⑪ Connect temperature probe and any other sensors to appropriate connections.

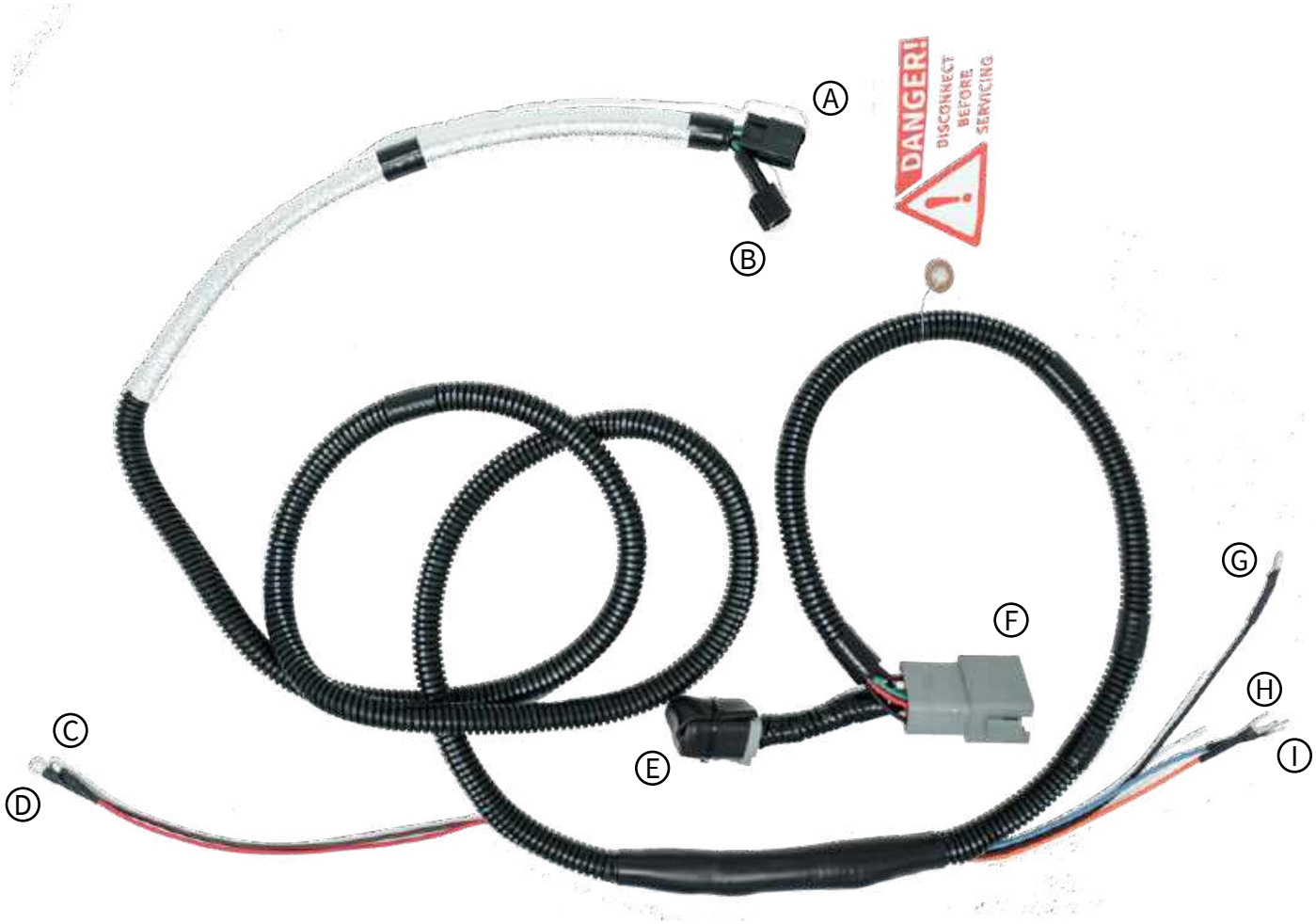


Make: Ochard Rite
Engine: Ford V-10
Autostart: Orchard Rite
Altrac Wiring Harness: WH112

Learn how to install the Altrac wiring harness for the engine below. Once installed, the Altrac Station will have access to power as well as readings from various sensors on the engine.



WH112



Key	Wire Color	Purpose	Terminal Type	Installation
A	Multiple	Speed Sensor (RPM)	2-Pin Female Furukawa Connector	RPM Splice
B	Multiple	Speed Sensor (RPM)	2-Pin Male Furukawa Connector	RPM Splice
C	Black	Ground	Size 10 Ring	Attach to wind machine ground.
D	Red	12VDC Power (Battery Voltage)	Size 10 Ring	Connect to 12VDC power on the wind machine.
E	Multiple	Accessory Port Connector	6-Pin Deutsch Connector	Plug in fuel sensors, soil moisture sensors, etc.
F	Multiple	Control Line Connector	12-Pin Deutsch Connector	Plug into the matching 12-pin connector on the ST100's control line.
G	Brown	Auto Switch Detection (12VDC)	Size 10 Ring	Connect to the 'Auto' side of the Manual-Off-Auto switch.
H	Orange	Power for Start Signal	Size 8 Fork	Replaces Orchard Rite temperature probe's black wire.
I	Blue	Start Signal	Size 8 Fork	Replaces Orchard Rite temperature probe's black wire.

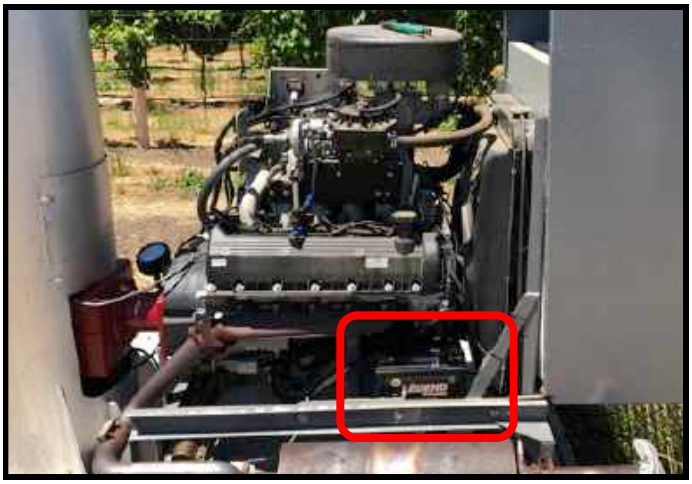
Tool List

Use	Tool
Wiring Harness Installation	Philips Screwdriver - Medium
Wiring Harness Installation	Philips Screwdriver - Large
Wiring Harness Installation	Flathead Screwdriver - Large
Wiring Harness Installation	Magnetic Bolt Retriever
Wiring Harness Installation	Channel Locks
Wiring Harness Installation	Electrical Tape
Wiring Harness Installation	4" Zip Ties
Wiring Harness Installation	12" Zip Ties
Wiring Harness Installation	18" Zip Ties
Wiring Harness Installation	Nippers (Flush Cutters)
Wiring Harness Installation	Adjustable Cresent Wrench
Wiring Harness Installation	Socket Drivers
Device Mounting	Impact Driver/Drill
Device Mounting	Metric Socket Set for Drill
Device Mounting	Drill Bits and Drivers
Device Mounting	Ladder
Device Mounting	Body Harness

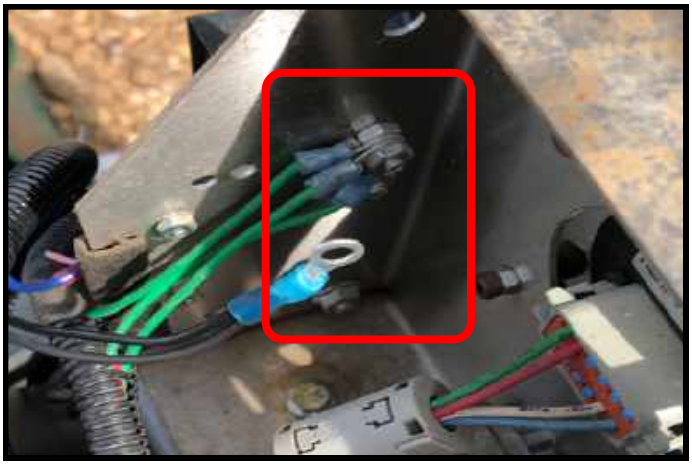


Before you attempt a wiring harness installation, you must unhook the wind machine battery and turn the Manual-Off-Auto switch to the OFF position.

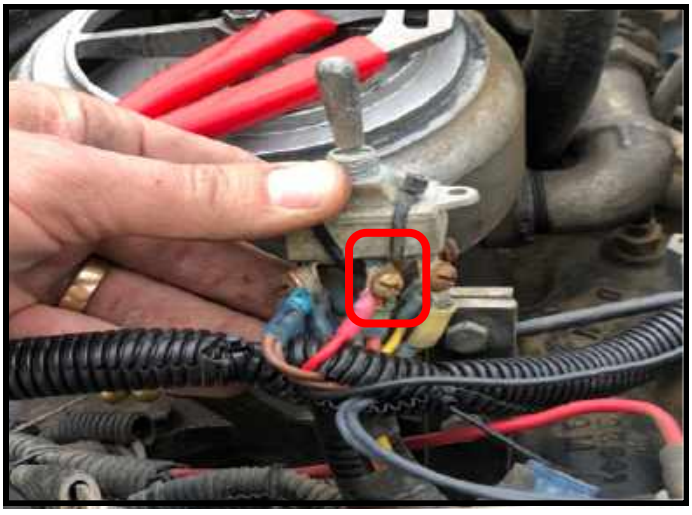
① Disconnect the wind machine engine's 12V battery.



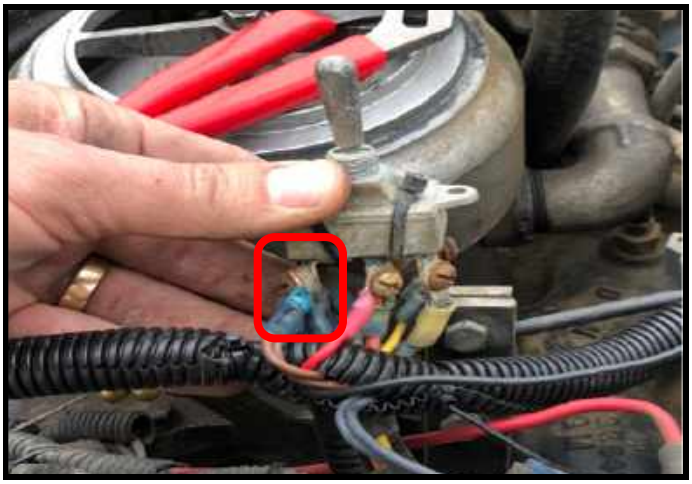
② Connect black ground wire to a ground source on engine. Other ground wires are usually green or black.



③ Connect Red Wire to the bottom-middle contact of the MAN-OFF-AUTO switch on the wind machine gauge panel. You will need to remove the switch with a crescent wrench.

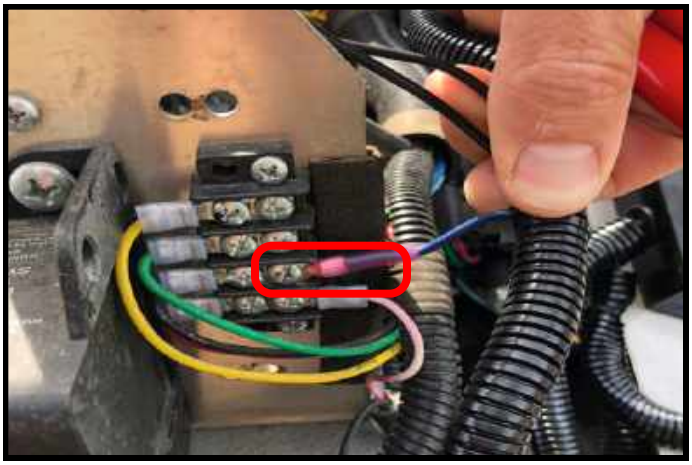


④ Connect Brown Wire to the bottom-left contact of the MAN-OFF-AUTO switch on the wind machine gauge panel. You will need to remove the switch with a crescent wrench.



⑤ Replace the black wire from Orchard Rite temperature probe with blue wire from harness.

HINT: Follow the temperature probe wire to the terminal block on the gauge panel to find the correct wires.



⑥ Replace red wire from Orchard Rite temperature probe with orange wire from harness.

HINT: Follow the temperature probe wire to the terminal block on the gauge panel to find the correct wires.



⑦ To attach the RPM sensor you will need to unplug the existing speed sensor connector from the sensor.

HINT: The sensor is located behind the wind machine battery near the fan belt. You will need to remove the wind machine battery to access the sensor.

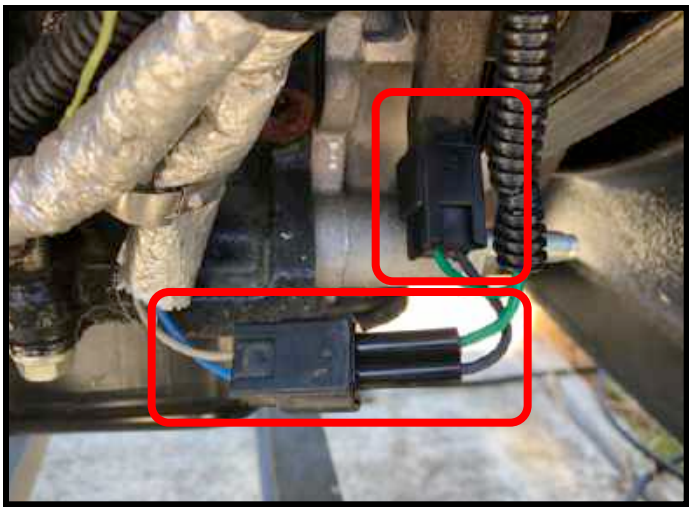


⑧ Locate the RPM connector pair on the Altrac Wiring Harness



⑨ Plug the female connector from the Altrac harness into the speed sensor and the original female connector into the male on the Altrac harness. This will create a bridge between the sensor and the original connector.

HINT: Secure the altrac wiring harness away from the fan belt and exhaust with zipties. The moving belt will quickly destroy wiring.



⑨ Reattach the wind machine battery.



⑩ Connect the new wiring harness to the Altrac Station via the 12-pin connector on the control line.

HINT: You should tuck the connector up under the wind machine hood when you are finished installing to maximize weather protection.



⑪ Connect temperature probe and any other sensors to appropriate connections.

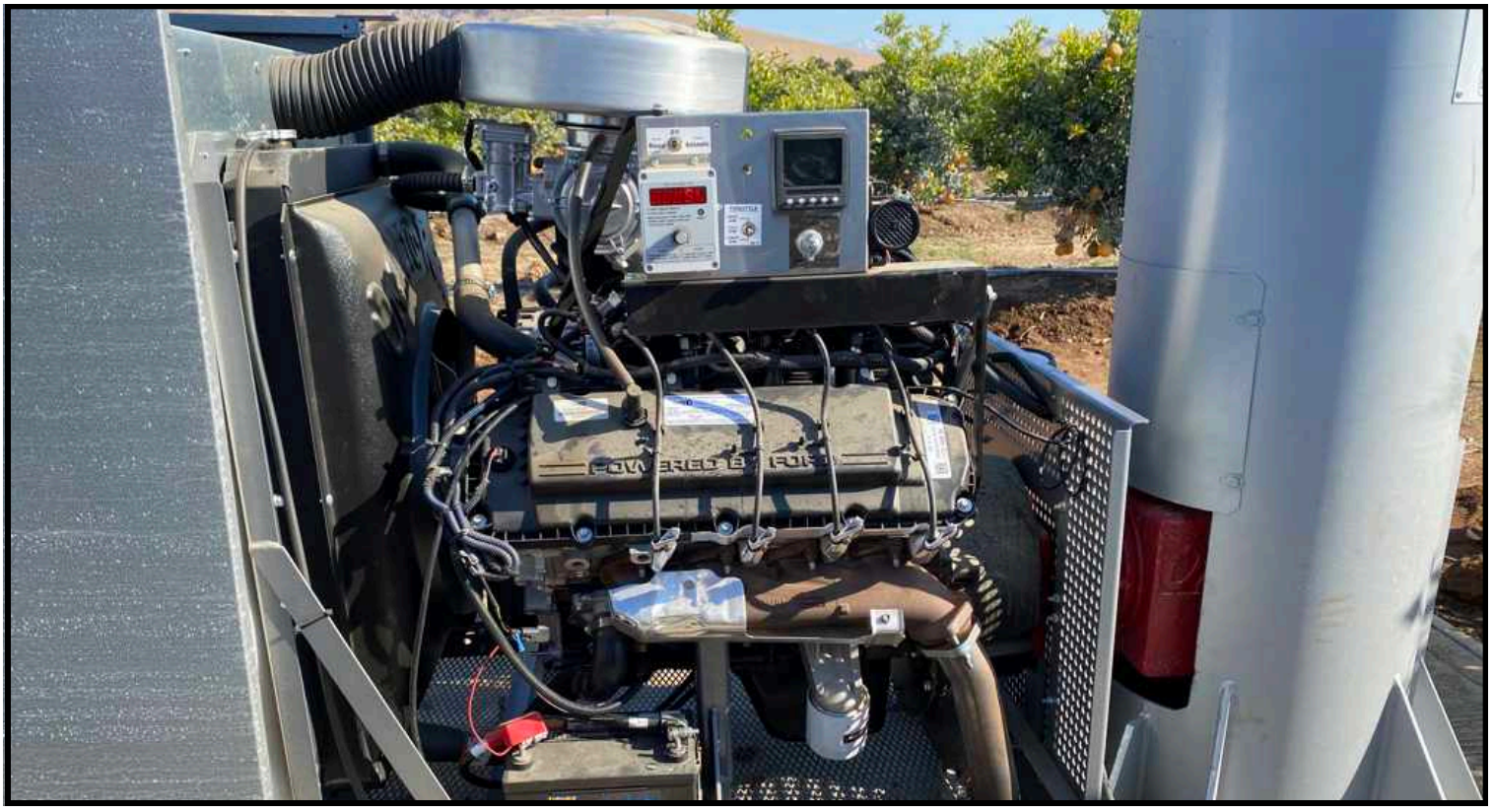


⑨ To run the wind machine you must have the Manual-Off-Auto switch in the 'Auto' position, and the autostart must be armed. You will see a blinking white light when the autostart is armed.

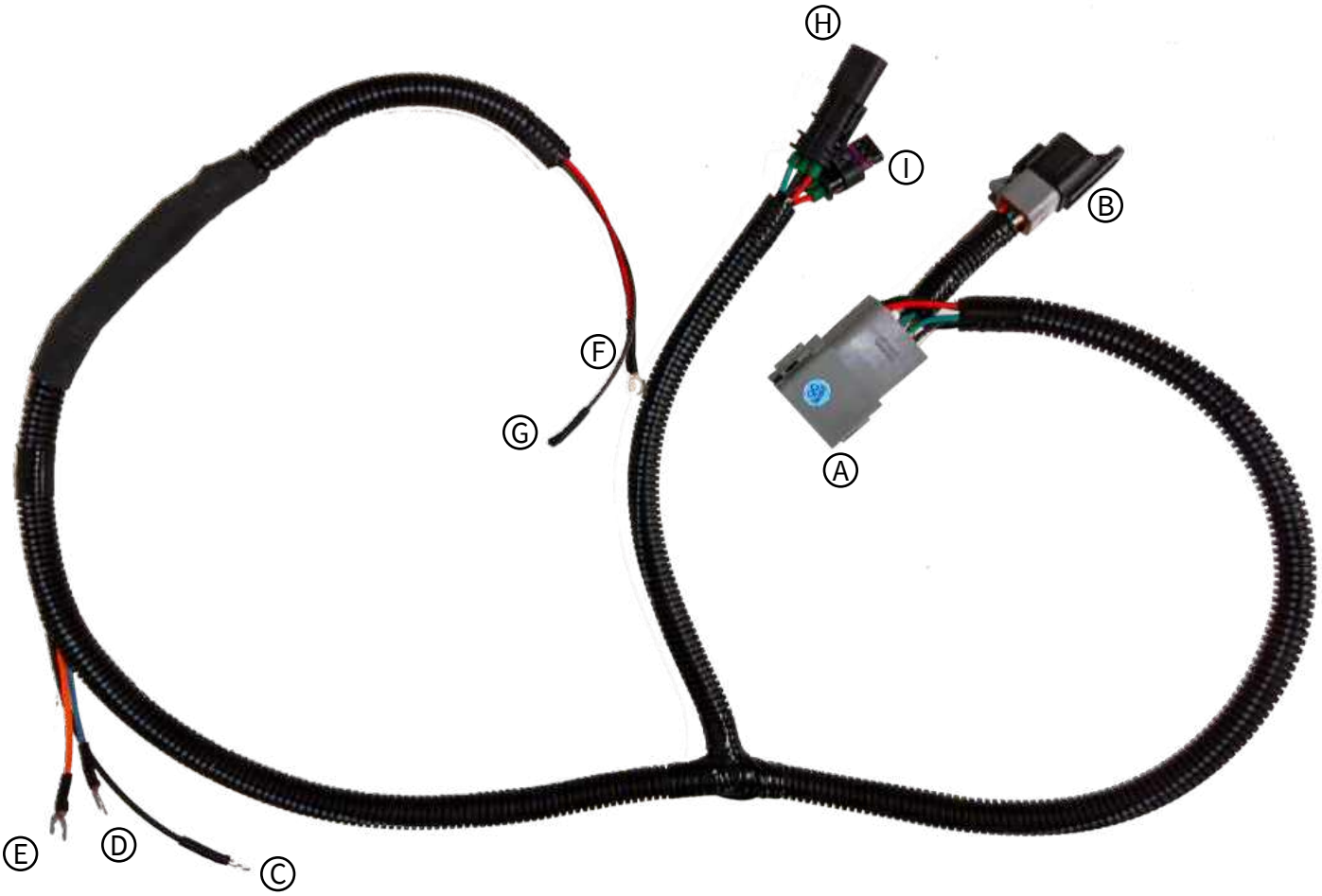


Make: Ochard Rite
Engine: Ford 6.2L V-8
Autostart: Orchard Rite
Altrac Wiring Harness: WH122

Learn how to install the Altrac wiring harness for the engine below. Once installed, the Altrac Station will have access to power as well as readings from various sensors on the engine.



WH122



Key	Wire Color	Purpose	Terminal Type	Installation
A	Multiple	Control Line Connector	12-Pin Deutsch Connector	Plug into the matching 12-pin connector on the ST100's control line.
B	Multiple	Accessory Port	6-Pin Deutsch Connector	Plug in fuel sensors, soil moisture sensors, etc.
C	Black	Ground	Size 10 Ring	Attach to wind machine ground.
D	Blue	Start Signal	Size 8 Fork	Replaces Orchard Rite temperature probe's black wire.
E	Orange	Power for Start Signal	Size 8 Fork	Replaces Orchard Rite temperature probe's black wire.
F	Red	Power	Size 10 Ring	Connect to 12VDC power on the wind machine.
G	Brown	Auto Position Detection	Size 10 Ring	Connect to the 'Auto' side of the Manual-Off-Auto switch.
H	Multiple	Speed Sensor (RPM)	Female	RPM Splice
I	Multiple	Speed Sensor (RPM)	Male	RPM Splice

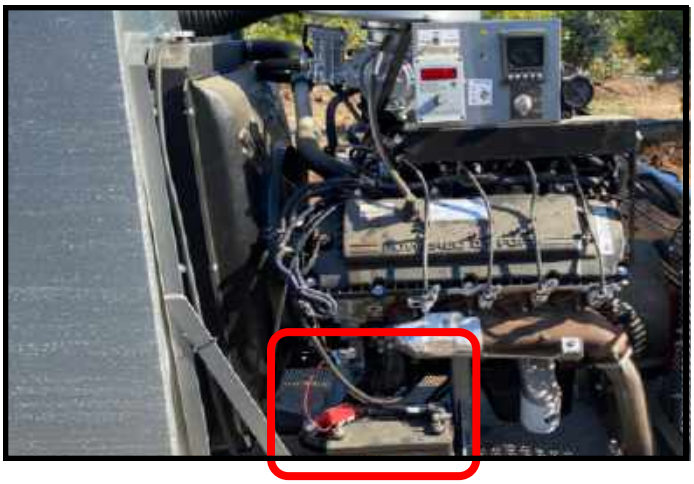
Tool List

Use	Tool
Wiring Harness Installation	Philips Screwdriver - Medium
Wiring Harness Installation	Philips Screwdriver - Large
Wiring Harness Installation	Flathead Screwdriver - Large
Wiring Harness Installation	Magnetic Bolt Retriever
Wiring Harness Installation	Channel Locks
Wiring Harness Installation	Electrical Tape
Wiring Harness Installation	4" Zip Ties
Wiring Harness Installation	12" Zip Ties
Wiring Harness Installation	18" Zip Ties
Wiring Harness Installation	Nippers (Flush Cutters)
Wiring Harness Installation	Adjustable Cresent Wrench
Wiring Harness Installation	Socket Drivers
Device Mounting	Impact Driver/Drill
Device Mounting	Metric Socket Set for Drill
Device Mounting	Drill Bits and Drivers
Device Mounting	Ladder
Device Mounting	Body Harness

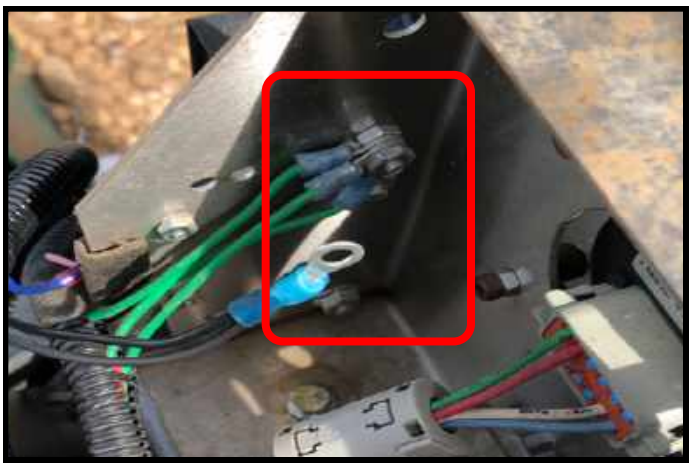


Before you attempt a wiring harness installation, you must unhook the wind machine battery and turn the Manual-Off-Auto switch to the OFF position.

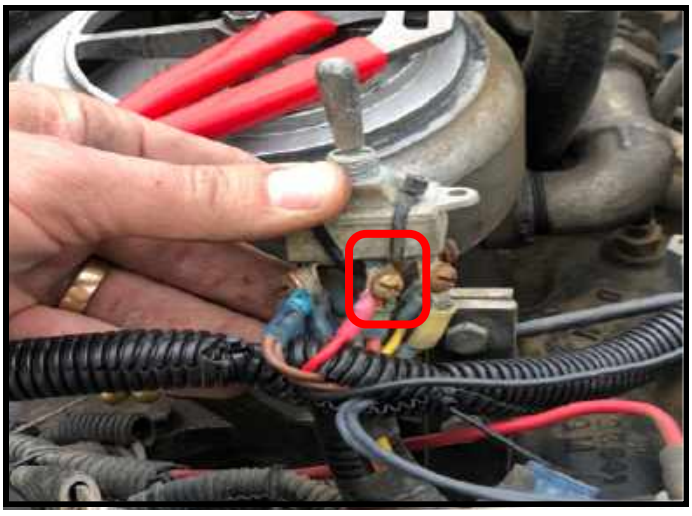
① Disconnect the wind machine engine's 12V battery.



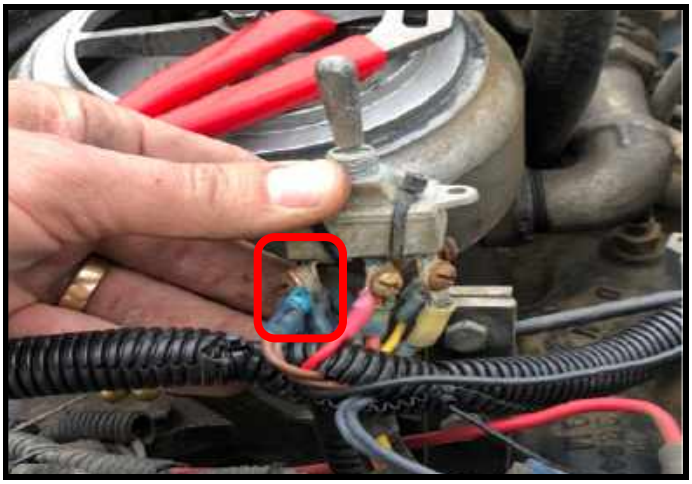
② Connect black ground wire to the ground souce on engine.



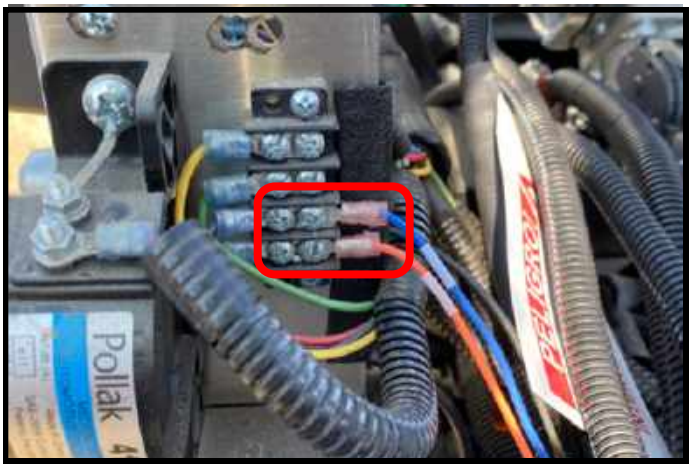
③ Connect Red Wire to the bottom-middle contact of the MAN-OFF-AUTO switch on the wind machine gauge panel. You will need to remove the switch with a crescent wrench.



④ Connect Brown Wire to the bottom-left contact of the MAN-OFF-AUTO switch on the wind machine gauge panel. You will need to remove the switch with a crescent wrench.



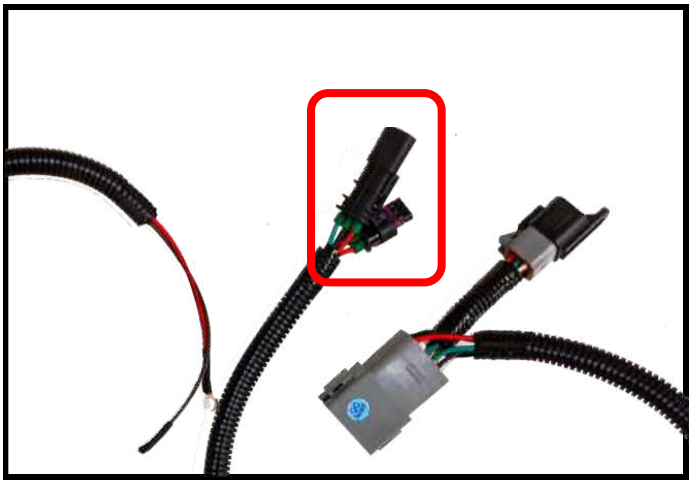
⑤ Replace the black wire from temperature probe with blue wire from harness. Replace red wire from temperature probe with orange wire from harness.



⑥ Locate the RPM sensor and Unplug the RPM Sensor connector from the sensor.



⑦ Locate the RPM connector pair on the Altrac Wiring Harness.



⑧ Plug in Female RPM Connector to the wind machine RPM sensor.



⑨ Plug in Male RPM Connector to the to the sensor wire.



⑩ Secure the altrac wiring harness away from the fan belt and exhaust.



⑪ Reattatch a battery.



⑫ Connect wiring harness to the control line.



⑪ Connect temperature probe and any other sensors to appropriate connections.



⑨ To run the wind machine you must have the Manual-Off-Auto switch in the ‘Auto’ position, and the autostart must be armed. You will see a blinking white light when the autostart is armed.

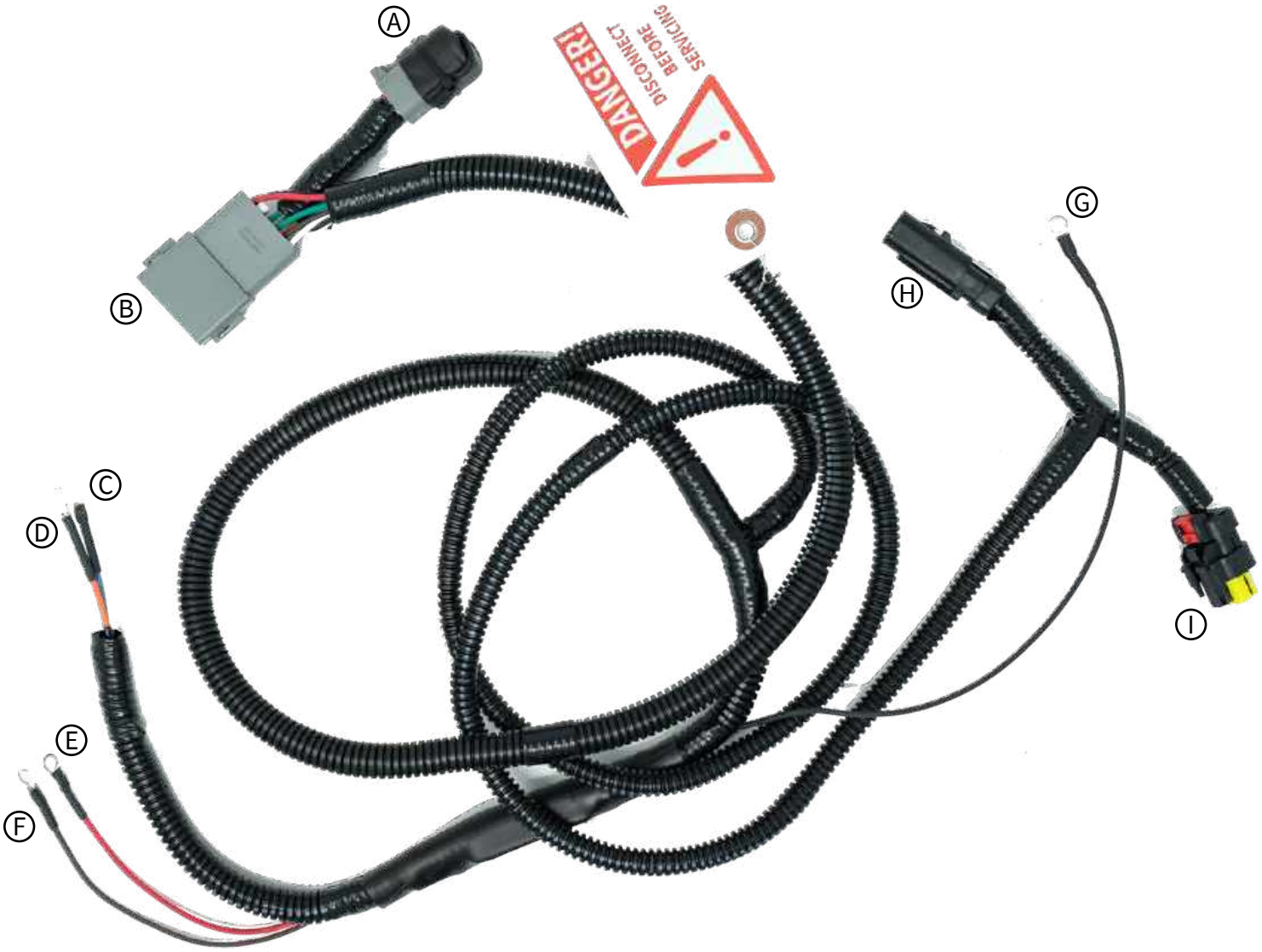


Make: Ochard Rite
Engine: CAT 4.4, 6.6, 7.1
Autostart: Orchard Rite (any model)
Altrac Wiring Harness: WH140

Learn how to install the Altrac wiring harness for the engine below. Once installed, the Altrac Station will have access to power as well as readings from various sensors on the engine.



WH140



Key	Wire Color	Purpose	Terminal Type	Installation
A	Multiple	Accessory Port	6-Pin Deutsch Connector	Plug in fuel sensors, soil moisture sensors, etc.
B	Multiple	Control Line Connector	12-Pin Deutsch Connector	Plug into the matching 12-pin connector on the ST100's control line.
C	Blue	Start Signal	SIze 8 Fork	Replaces Orchard Rite temperature probe's black wire.
D	Orange	Power for Start Signal	Size 8 Fork	Replaces Orchard Rite temperature probe's black wire.
E	Red	Battery Voltage (12VDC)	Size 10 Ring	Connect to 12VDC power on the wind machine.
F	Brown	Auto Switch Detection (12VDC)	Size 10 Ring	Connect to the 'Auto' side of the Manual-Off-Auto switch.
G	Black	Ground	Size 10 Ring	Attach to wind machine ground.
H	Multiple	Speed Sensor (RPM)	2-Pin Female AMPSeal 16	Splice into the RMP Signal
I	Multiple	Speed Sensor (RPM)	2-Pin Male AMPSeal 16	Splice into the RMP Signal

Tool List

Use	Tool
Wiring Harness Installation	Philips Screwdriver - Medium
Wiring Harness Installation	Philips Screwdriver - Large
Wiring Harness Installation	Flathead Screwdriver - Large
Wiring Harness Installation	Magnetic Bolt Retriever
Wiring Harness Installation	Channel Locks
Wiring Harness Installation	Electrical Tape
Wiring Harness Installation	4" Zip Ties
Wiring Harness Installation	12" Zip Ties
Wiring Harness Installation	18" Zip Ties
Wiring Harness Installation	Nippers (Flush Cutters)
Wiring Harness Installation	Adjustable Cresent Wrench
Wiring Harness Installation	Socket Drivers
Device Mounting	Impact Driver/Drill
Device Mounting	Metric Socket Set for Drill
Device Mounting	Drill Bits and Drivers
Device Mounting	Ladder
Device Mounting	Body Harness

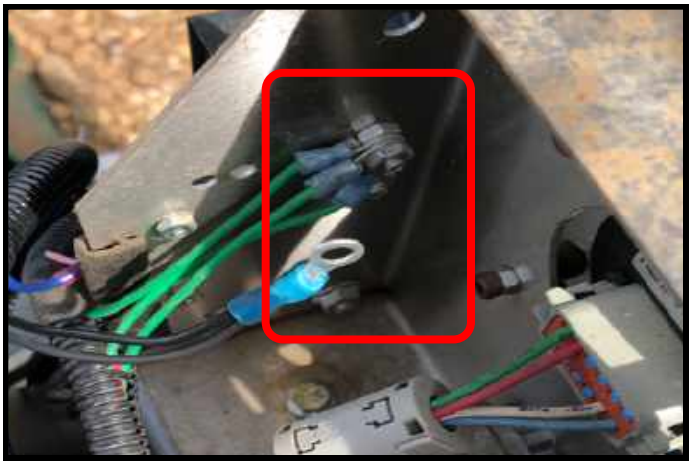


Before you attempt a wiring harness installation, you must unhook the wind machine battery and turn the Manual-Off-Auto switch to the OFF position.

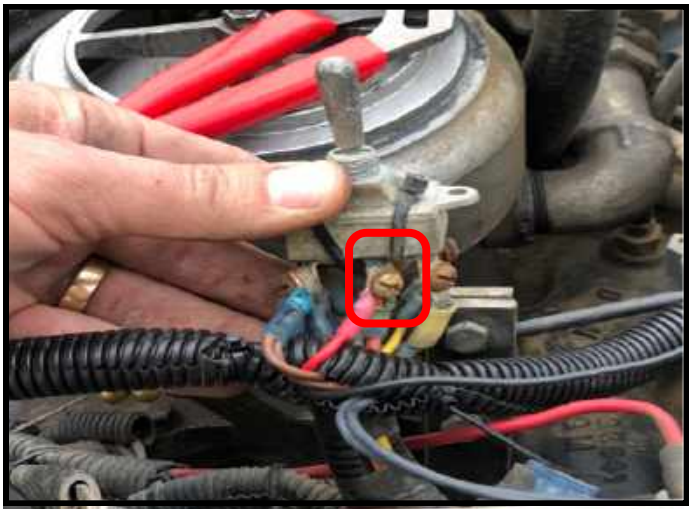
① Disconnect the wind machine engine's 12V battery.



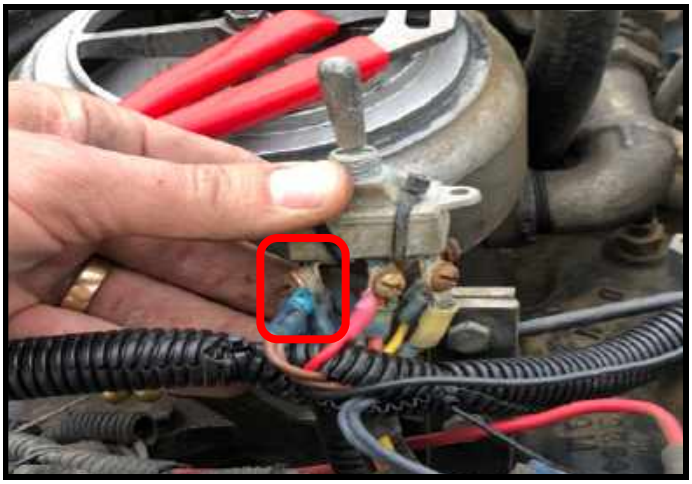
② Connect black ground wire to a ground source on engine. Other ground wires are usually green or black.



③ Connect Red Wire to the bottom-middle contact of the MAN-OFF-AUTO switch on the wind machine gauge panel. You will need to remove the switch with a crescent wrench.

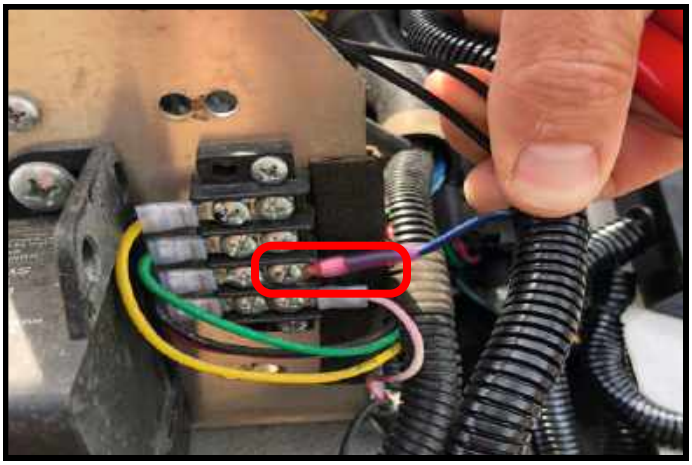


④ Connect Brown Wire to the bottom-left contact of the MAN-OFF-AUTO switch on the wind machine gauge panel. You will need to remove the switch with a crescent wrench.



⑤ Replace the black wire from Orchard Rite temperature probe with blue wire from harness.

HINT: Follow the temperature probe wire to the terminal block on the gauge panel to find the correct wires.



⑥ Replace red wire from Orchard Rite temperature probe with orange wire from harness.

HINT: Follow the temperature probe wire to the terminal block on the gauge panel to find the correct wires.



⑦ Find the RPM speed sensor. On CAT engine, the speed sensor is located directly behind or near the fuel filters on the right side of the engine (depending on your engine type).



⑧ To attach the RPM sensor you will need to unplug the existing speed sensor connector from the sensor.



⑨ Plug the female connector from the Altrac harness into the speed sensor and the original female connector into the male on the Altrac harness. This will create a bridge between the sensor and the original connector.



⑨ Reattach the wind machine battery.



⑩ Connect the new wiring harness to the Altrac Station via the 12-pin connector on the control line.

HINT: You should tuck the connector up under the wind machine hood when you are finished installing to maximize weather protection.



⑪ Connect temperature probe and any other sensors to appropriate connections.



⑨ To run the wind machine you must have the Manual-Off-Auto switch in the 'Auto' position, and the autostart must be armed. You will see a blinking white light when the autostart is armed.

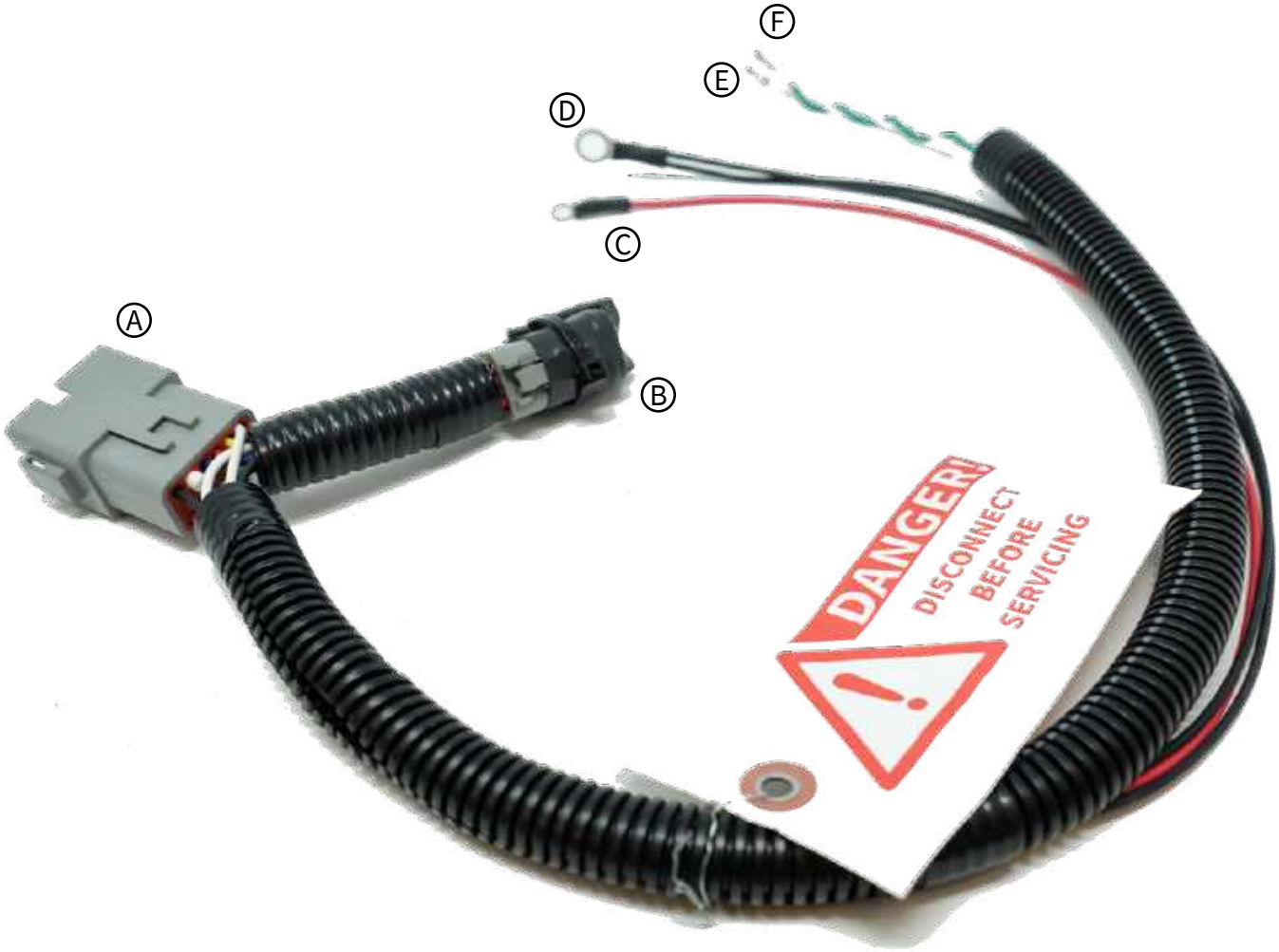


Make: Amarillo
Engine: Ford V-10 and Ford 6.2L V-8
Autostart: Murphy MPC-20
Altrac Wiring Harness: WH110

Learn how to install the Altrac wiring harness for the engine below. Once installed, the Altrac Station will have access to power as well as readings from various sensors on the engine.



WH110



Key	Wire Color	Purpose	Terminal Type	Installation
A	Multiple	Control Line Connector	12-Pin Deutsch Connector	Plug into the matching 12-pin connector on the ST100's control line.
B	Multiple	Accessory Port	6-Pin Deutsch Connector	Plug in fuel sensors, soil moisture sensors, etc.
C	Red	Battery Voltage (12VDC)	Size 10 Ring	Connect to 12VDC power on the wind machine.
D	Black	Ground	Size 10 Ring	Attach to wind machine ground.
E	Green	Autostart Communication	MPC-20 90 Pin	Install in the Murphy MPC-20 Connector.
F	White	Autostart Communication	MPC-20 90 Pin	Install in the Murphy MPC-20 Connector.

Tool List

Use	Tool
Wiring Harness Installation	Philips Screwdriver - Medium
Wiring Harness Installation	Philips Screwdriver - Large
Wiring Harness Installation	Flathead Screwdriver - Large
Wiring Harness Installation	Magnetic Bolt Retriever
Wiring Harness Installation	Electrical Tape
Wiring Harness Installation	4" Zip Ties
Wiring Harness Installation	12" Zip Ties
Wiring Harness Installation	18" Zip Ties
Device Mounting	Impact Driver/Drill
Device Mounting	Metric Socket Set for Drill
Device Mounting	Drill Bits and Drivers
Device Mounting	Ladder
Device Mounting	Body Harness



Before you attempt a wiring harness installation, you must unhook the wind machine battery and turn the Manual-Off-Auto switch to the OFF position.

① Disconnect the wind machine battery.



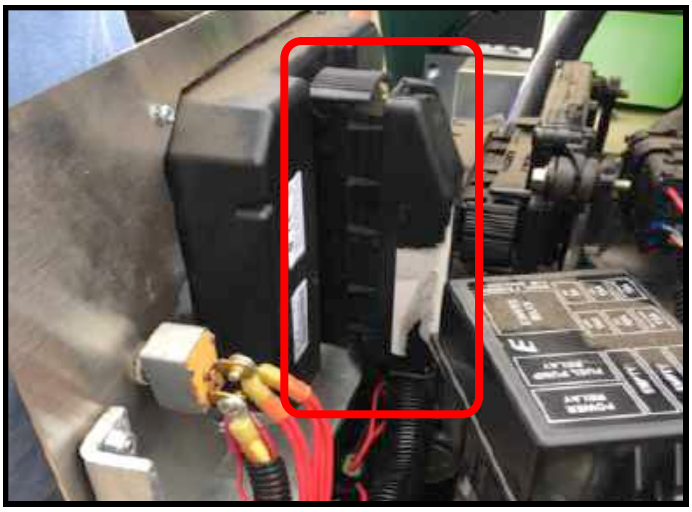
② Connect black ground wire to the ground source on engine.



③ Connect Red 12v Power wire to the ring terminal.



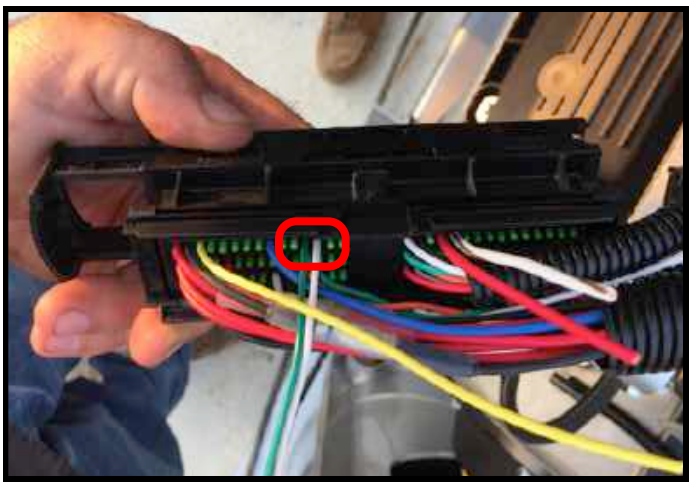
④ Locate MPC-20 Connector and remove. You will need to pull up on the top grip to detach the connector.



⑤ Once removed, you will need to detach the yellow wedgelock from the face of the connector. Gently lift on the tabs to allow full removal of the yellow wedgelock.



⑥ Insert the green wire to pin 12 and connect the white wire to pin 13 on the MPC-20 connector.



⑦ Close connector and reconnect to MPC-20.



⑧ Reattach the wind machine battery.



⑨ Turn on the MPC-20 and Update the firmware. Turn off the MPC-20 when completed.

For more information on updating your firmware, please contact support@altrac.io.



⑩ Connect the new wiring harness to the Altrac Station via the 12-pin connector on the control line.

HINT: You should tuck the connector up under the wind machine hood when you are finished installing to maximize weather protection.



⑪ Connect temperature probe and any other sensors to appropriate connections.

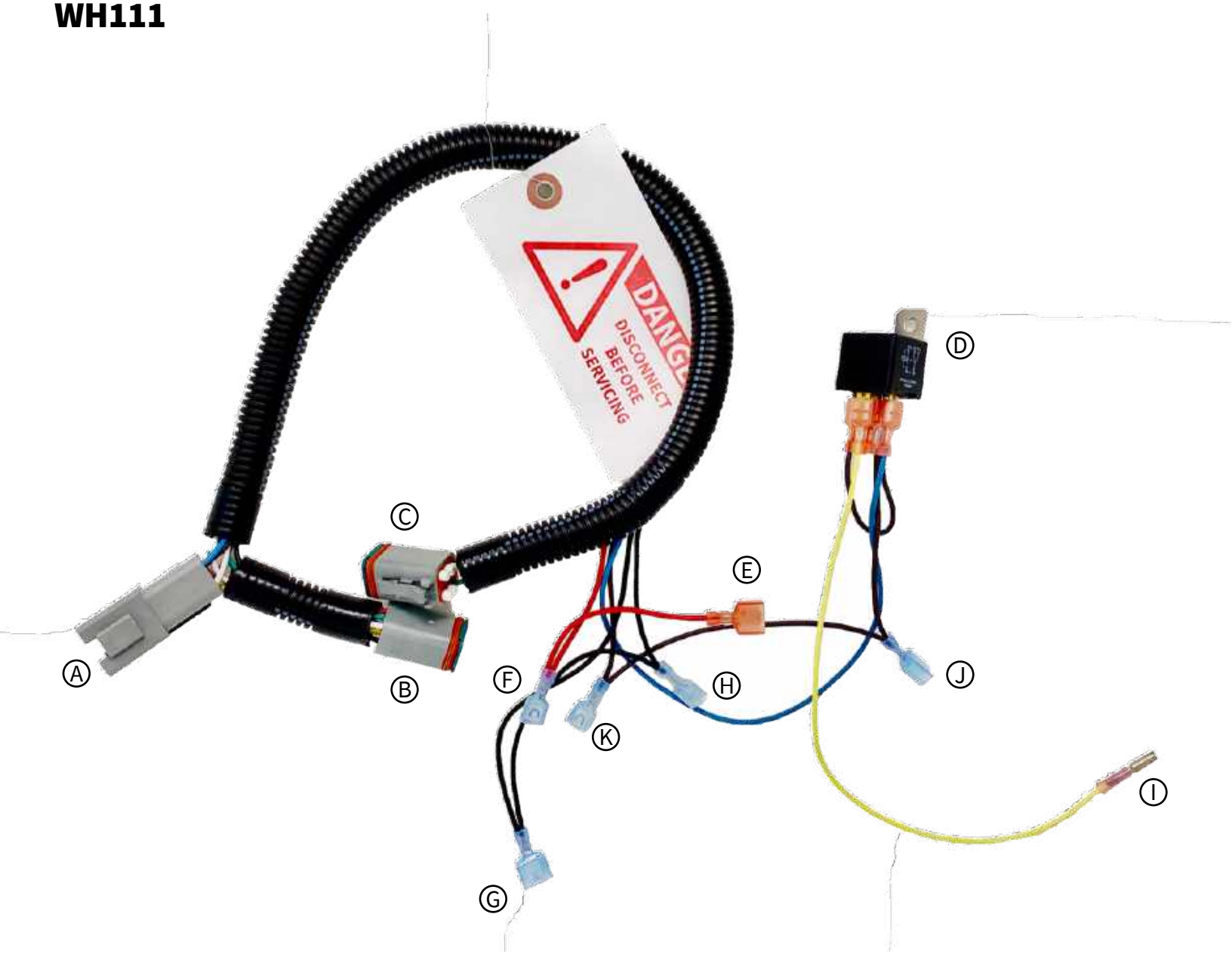


Make: Chinook/HF Hauff
Engine: Ford V-10
Autostart: Multiple, PV101 for Gauges
Altrac Wiring Harness: WH111

Learn how to install the Altrac wiring harness for the engine below. Once installed, the Altrac Station will have access to power as well as readings from various sensors on the engine.



WH111



Key	Wire Color	Purpose	Terminal Type	Installation
A	Multiple	Control Line Connector	12-Pin Deutsch Connector	Plug into the matching 12-pin connector on the ST100's control line.
B	Multiple	Accessory Port	6-Pin Deutsch Connector	Attach additional sensors such as fuel level or soil mositure.
C	Multiple	PV101 Connection	6-Pin Deutsch Connector	Attach to port B on the PV101 gauge panel.
D	Multiple	Controls the wind machine.	Relay	Mount near the gauge panel with a self tapping screw.
E	Red	Battery Voltage (12VDC)	Quick Connect Male	Connect to bottom-middle contact on Manual-Off-Auto switch (red wire).
F	Red	Battery Voltage (12VDC)	Quick Connect Female	Connect to bottom-middle contact on Manual-Off-Auto switch (red wire).
G	Black	Ground	Quick Connect Male	Connect to middle contact on RPM rocker switch (green wire).
H	Black	Ground	Quick Connect Female	Connect to middle contact on RPM rocker switch (green wire).
I	Yellow	Call to Start	5mm Bullet Connector	Insert into the trailer connector (matching yellow wire).
J	Brown	Auto Position Detection	Quick Connect Male	Connect to the 'Auto' side of the Manual-Off-Auto switch (brown wire).
K	Brown	Auto Position Detection	Quick Connect Female	Connect to the 'Auto' side of the Manual-Off-Auto switch (brown wire).

Tool List

Use	Tool
Wiring Harness Installation	Philips Screwdriver - Medium
Wiring Harness Installation	Philips Screwdriver - Large
Wiring Harness Installation	Flathead Screwdriver - Large
Wiring Harness Installation	Magnetic Bolt Retriever
Wiring Harness Installation	Electrical Tape
Wiring Harness Installation	4" Zip Ties
Wiring Harness Installation	12" Zip Ties
Wiring Harness Installation	18" Zip Ties
Device Mounting	Impact Driver/Drill
Device Mounting	Metric Socket Set for Drill
Device Mounting	Drill Bits and Drivers
Device Mounting	Ladder
Device Mounting	Body Harness

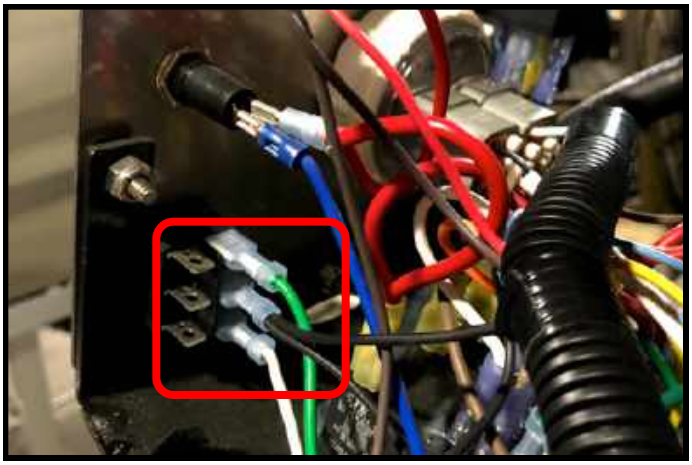


Before you attempt a wiring harness installation, you must unhook the wind machine battery and turn the Manual-Off-Auto switch to the OFF position.

① Disconnect the wind machine 12V battery.



② Connect the black ground wires to the ground source on engine. Remove the green wire from the middle contact on the RPM rocker switch and jump it with the two black wires on the Altrac wiring harness.



③ Connect red power wire to bottom-middle contact on the Manual-Off-Auto switch. Remove the red wire and jump it with the two red wires on the Altrac wiring harness.



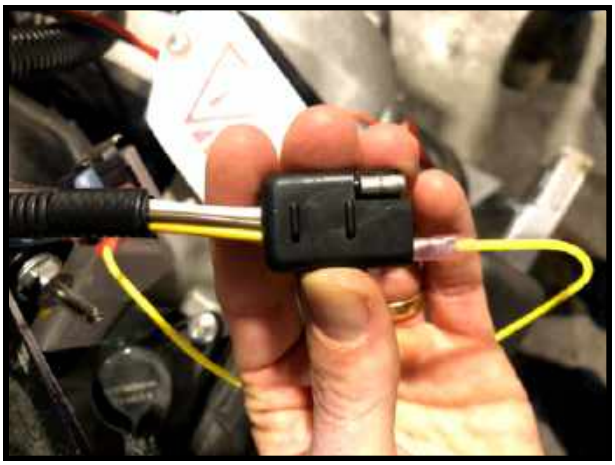
④ Connect the brown auto position detect wire to the auto side (left) of the Manual-Off-Auto switch. Remove the brown wire from the auto side of the switch and jump it with the two brown wires on the Altrac wiring harness.



⑤ Connect the grey 6-pin connector to the open slot 'B' on the back of the PV101.



⑥ Connect the yellow call to start wire to the yellow wire in the tailer connector on the wind machine engine. This trailer connector is usually plugged into the exisiting temperature probe controller.



⑦ You must program the PV101 to send gauge readings to the Altrac Station. To start, turn the Manual-Off-Auto switch to Manual to power up the PV101.



⑦ Press the Menu button (three dots + lines) and select ‘Utilities’.



⑧ Select ‘OEM’.



⑨ Enter Password.



⑩ Select ‘ModBus Setup’.



⑪ Select ‘Slave Active’. Once you have selected ‘Slave Active’ you can close the menu and return to the main page.

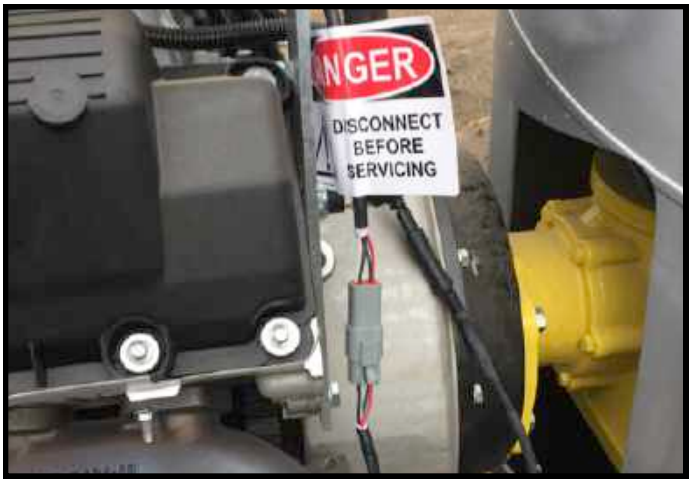


⑨ Reattach the wind machine battery.



⑩ Connect the new wiring harness to the Altrac Station via the 12-pin connector on the control line.

HINT: You should tuck the connector up under the wind machine hood when you are finished installing to maximize weather protection.



⑪ Connect temperature probe and any other sensors to appropriate connections.

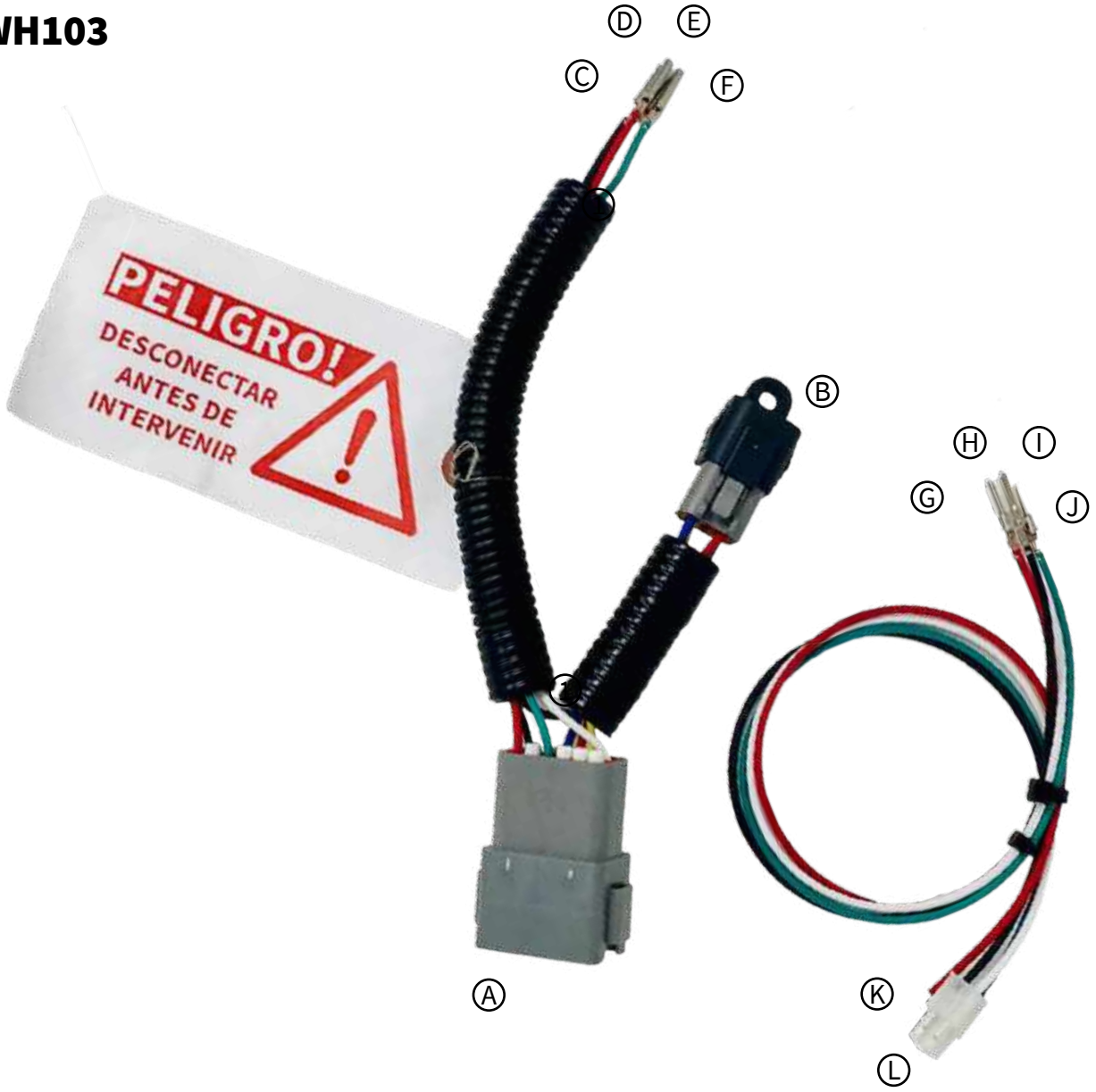


Make: Vamco
Engine: 360, 454, 460
Autostart: Murphy Prolite
Altrac Wiring Harness: WH103

Learn how to install the Altrac wiring harness for the engine below. Once installed, the Altrac Station will have access to power as well as readings from various sensors on the engine.



WH103



Key	Wire Color	Purpose	Terminal Type	Installation
A	Multiple	Control Line Connector	12-Pin Deutsch Connector	Plug into the matching 12-pin connector on the ST100's control line.
B	Multiple	Accessory Port	6-Pin Deutsch Connector	Plug in fuel sensors, soil moisture sensors, etc.
C	Red	Power (12VDC)	Size 16 Contact Deutsch Terminal - Female	Insert into position 'K' on the circular connector on the exterior of the autostart.
D	Black	Ground	Size 16 Contact Deutsch Terminal - Female	Insert into position 'M' on the circular connector on the exterior of the autostart.
E	White	ModBus Communication (RS485A)	Size 16 Contact Deutsch Terminal - Female	Insert into position 'N' on tthe circular connector on the exterior of the autostart.
F	Green	ModBus Communication (RS485B)	Size 16 Contact Deutsch Terminal - Female	Insert into position 'P' on the circular connector on the exterior of the autostart.
G	Red	Power (12VDC)	Size 16 Contact Deutsch Terminal - Male	Insert into position 'K' on the circular connector on the interior of the autostart.
H	Black	Ground	Size 16 Contact Deutsch Terminal - Male	Insert into position 'M' on tthe circular connector on the interior of the autostart.
I	White	ModBus Communication (RS485A)	Size 16 Contact Deutsch Terminal - Male	Insert into position 'N' on tthe circular connector on the interior of the autostart.
J	Green	ModBus Communication (RS485B)	Size 16 Contact Deutsch Terminal - Male	Insert into position 'P' on the circular connector on the interior of the autostart.
K	Red	Power (12VDC)	Stripped 18AWG GXL Wire	Clip into the wago connector supplying 12VDC power on interior of enclosure.
L	Multiple	Connects all I/O to Autostart	Molex Mini Pin Jr. 4 Position	Clip into the mating molex connector on the bottom right of the Prolite PCB.

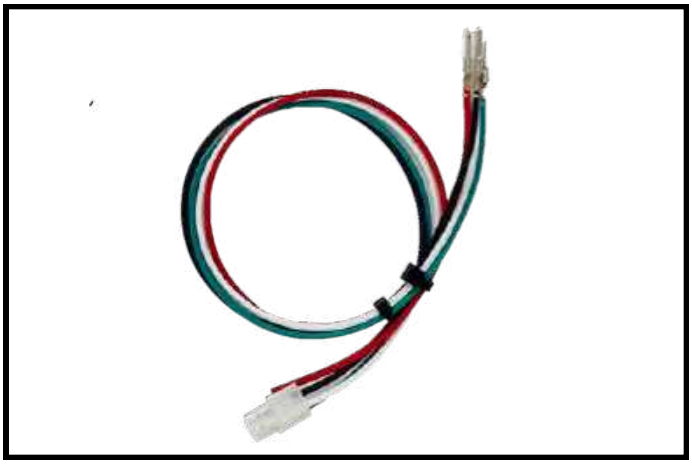
① Detach battery cables from the wind machine battery.



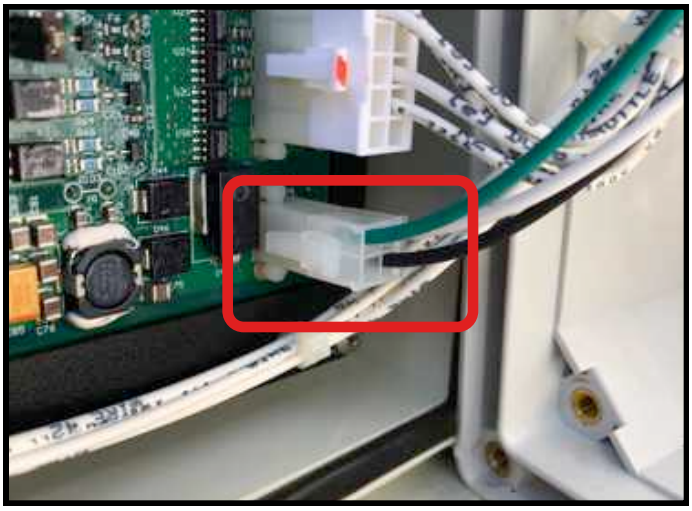
② Open the prolite controller by unscrewing the four screws in each corner of the enclosure.



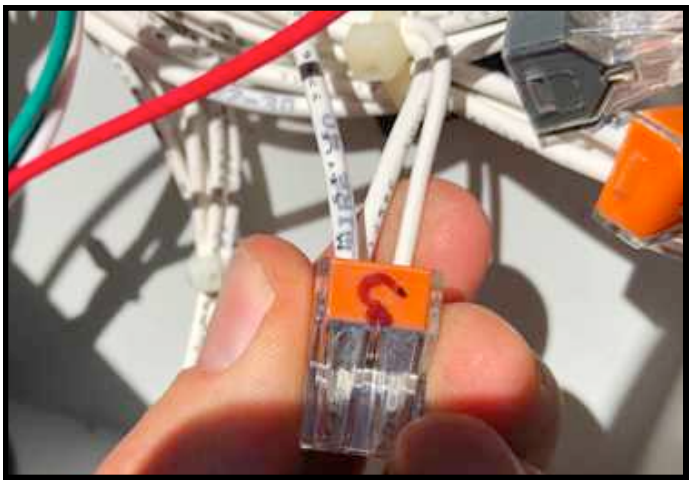
③ Find the internal wiring harness for the Prolite.



④ On the Prolite circuit board, plug in the Altrac wiring harness's molex connector into the 4-pin molex in the bottom right corner of the PCB.

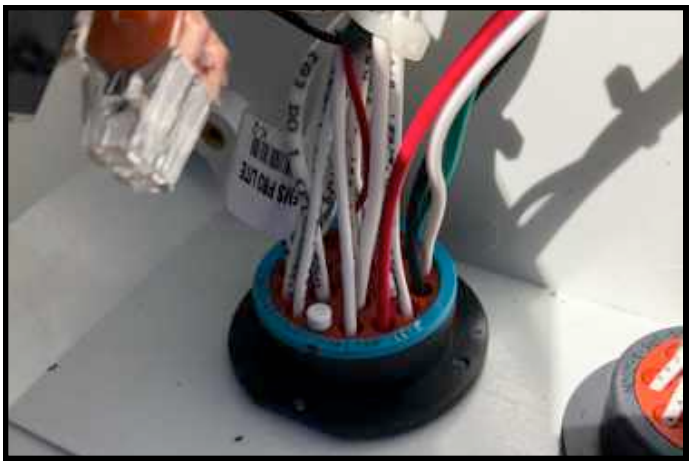


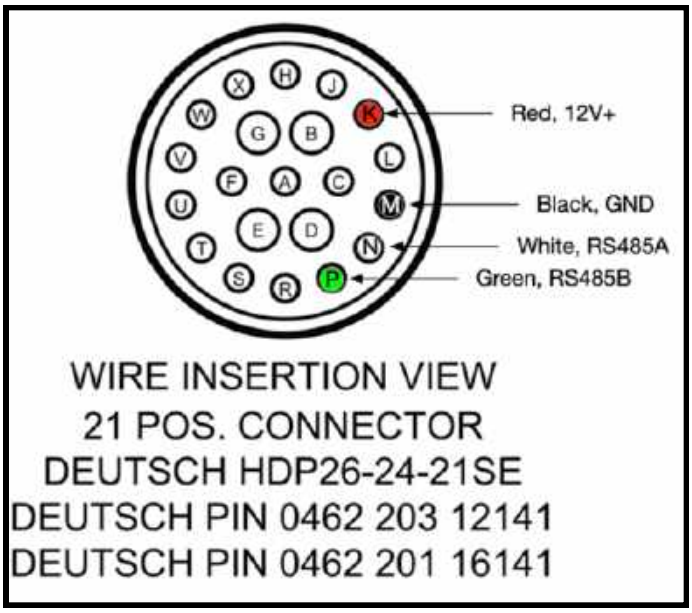
⑤ Plug in the stripped red wire (K) into the spring clamp labeled 3.



⑥ On the bottom of the enclosure interior, plug the four wires (G, H, I, J) with male pins into the 21-pin connector. They will lock in place.

Red	→	K
Black	→	M
White	→	N
Green	→	P





⑦ Close the Prolite and replace screws.



⑧ Find Altrac's external wiring harness for the Prolite.



⑨ On the Prolite's exterior plug the four wires (C, D, E, F) with female pins attached into the 21-pin connector. They will lock in place.

Red	→	K
Black	→	M
White	→	N
Green	→	P



⑩ Reattach the wind machine battery.



⑪ Connect the wiring harness to the contol line.



⑫ Connect temperature probe and any other sensors to appropriate connections.

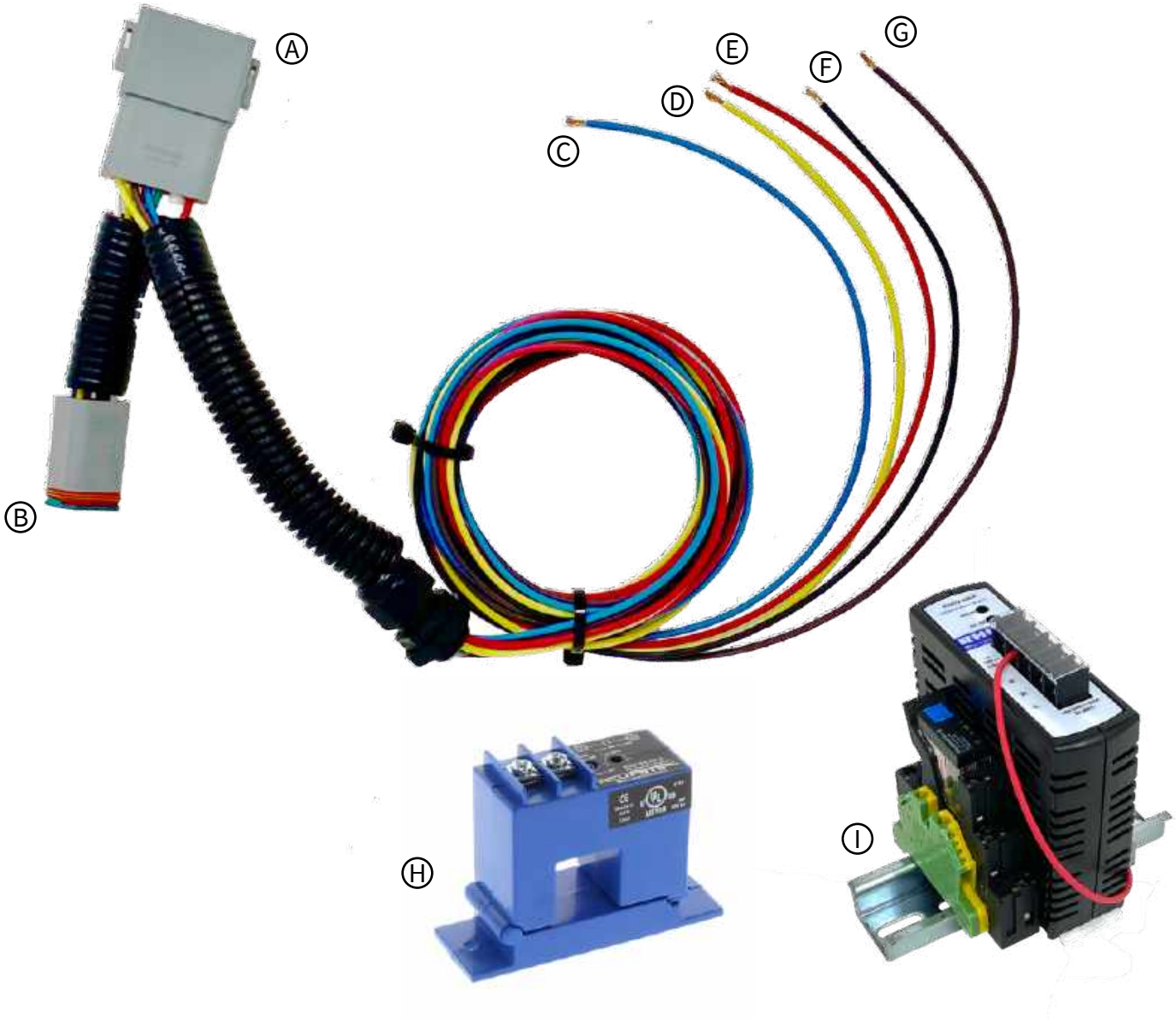


Make: Any
Engine: Electric
Autostart: Thermostat
Altrac Wiring Harness: WH121

Learn how to install the Altrac wiring harness for the engine below. Once installed, the Altrac Station will have access to power as well as readings from various sensors on the engine.



WH121

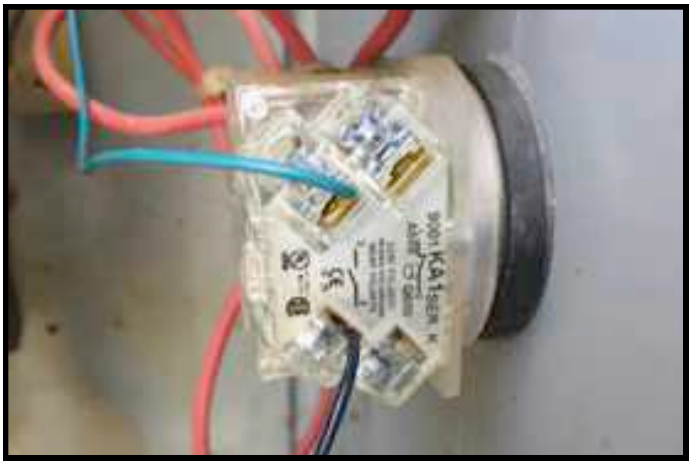


Key	Wire Color	Purpose	Terminal Type	Installation
A	Multiple	Control Line Connector	12-Pin Deutsch Connector	Plug into the matching 12-pin connector on the ST100's control line.
B	Multiple	Accessory Port	6-Pin Deutsch Connector	Plug in fuel sensors, soil moisture sensors, etc.
C	Blue	Current Switch	18AWG GXL Wire	Wire to current switch.
D	Yellow	Call to Start	18AWG GXL Wire	Wire to relay.
E	Red	Power	18AWG GXL Wire	Wire to power supply.
F	Black	Ground	18AWG GXL Wire	Attach to ground.
G	Brown	Auto Position Detection	18AWG GXL Wire	Wire to auto side of the Manual-Off-Auto switch.
H	None	Detects when the wind machine is running	Current Switch	Pass current carrying cable through current switch.
I	None	Powers Altrac Station and other circuits	Power Supply	Install power supply inside the cabinet.

① Install the 12VDC power supply in the electrical cabinet so that the Altrac Station receives power. Connect 12V positive to the red wire and ground to the black wire on the WH121 wiring harness.



② Add a new contact block to the switch on the wind machine, so the Altrac knows when the wind machine is in AUTO. The brown wire of the wiring harness should have 12VDC power when switch is in the auto position.

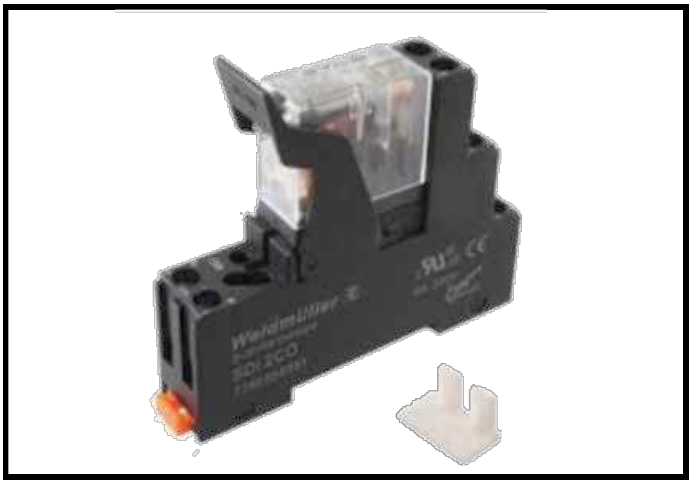


③ Remove one of the legs sending power to the motor and pass through the current switch. Connect the blue wire of the wiring harness to the Normally Open side of the sensor.

The current switch may need adjustment. Refer to included current switch user manual.



④ Install a relay to turn the wind machine on/off. The coil needs to be rated for 12VDC. The contacts need to be rated for the contactor coil on the wind machine, normally 250VAC. Connect the yellow wire on the WH121 to the '-' side of the coil.



Rated switching voltage	250 V AC
Max. switching voltage, DC	250 V
AC switching capacity (resistive), max.	2500 VA
Switch-on delay	≤ 15 ms
Min. switching power	10 mA @ 12 V, 100 mA @ 5 V

Max. switching voltage, AC	250 V
Continuous current	10 A
DC switching capacity (resistive), max.	240 W @ 24 V
Switch-off delay	≤ 10 ms
Max. switching frequency at rated load	0.1 Hz

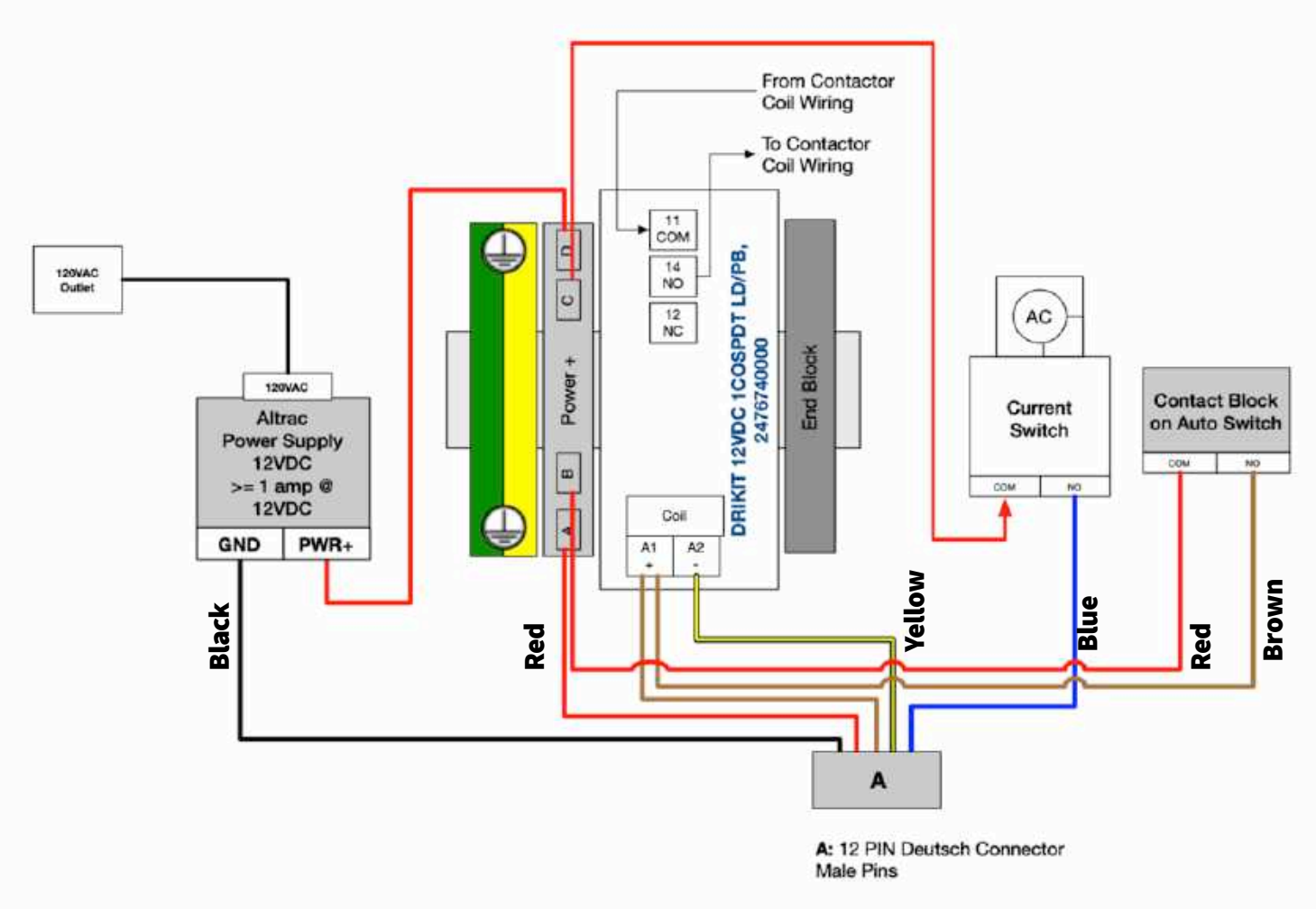
Schematic

NOTE: Connecting the Altrac equipment should only be done by a licensed electrician following local codes. Improper installation could result in shock or fire hazard.

Relay Requirements:

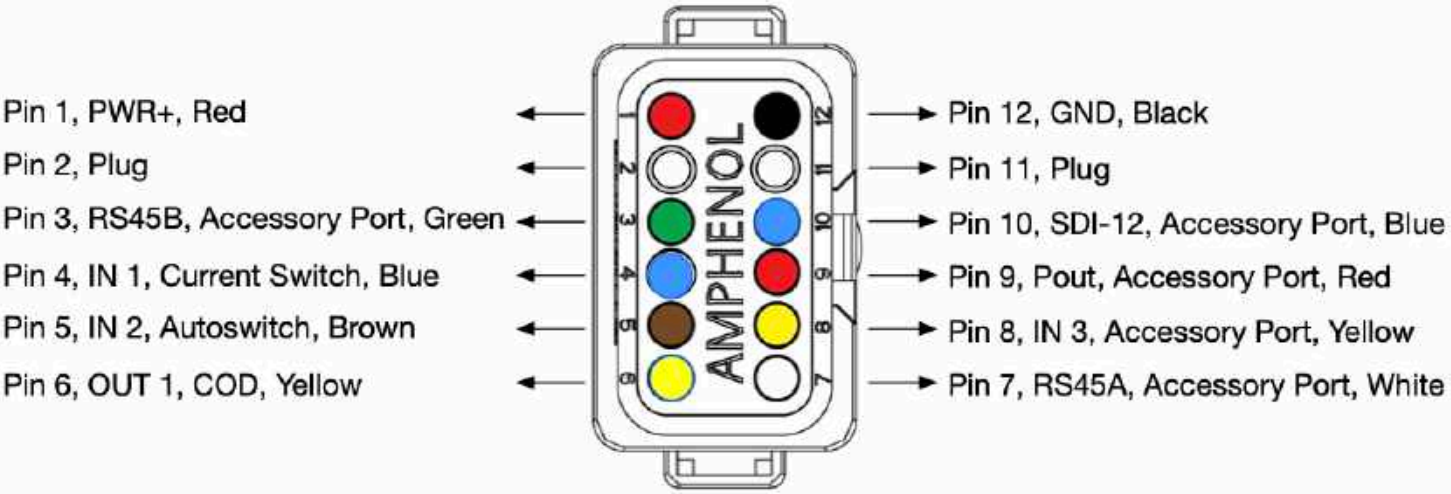
- Coil rated for 12VDC
- Contacts rated for voltage of Wind Machine contactor coil (Usually 250VAC)

NOTE: If you have a delay timer, wire the relay in before the timer, so the timer still functions.



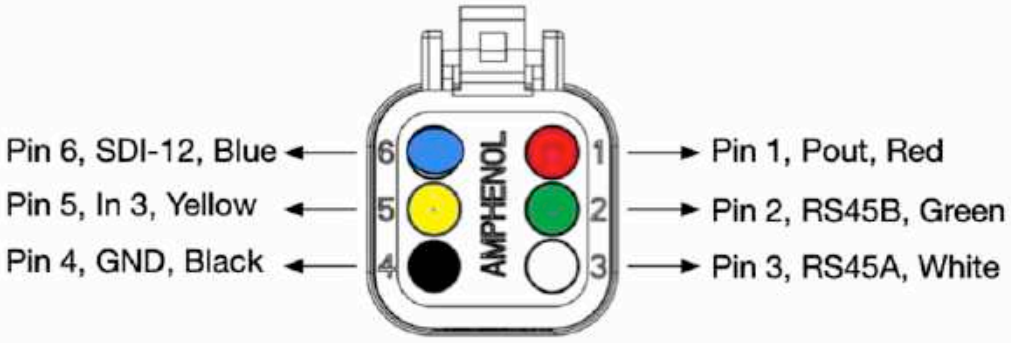
Schematic

A
12 PIN Deutsch Connector - Male Pins



View from Back

6 PIN Deutsch Connector - Female Pins
Fuel Sensor



View from Back

Operating Instructions - Web App

Log in to your Altrac account. To set up an account, please contact the distributor who installed your device or support@altrac.io.



Operation Graph
The wind machine operation graph visually shows when the pump is running (green) or when the wind machine is off but is scheduled to run (red). You can change the default settings by clicking the settings icon in the top right.

Controller State
Arm your wind machine by turning the Controller State to AUTO. Disarm the wind machine with MAN (manual).

Temperature
The temperature tile shows the current temperature reading. Click on tile to see the associated history and graph.

Battery Voltage
The battery voltage tile shows the current voltage reading. Click on tile to see the associated history and graph.

Settings
Click the settings icon on the device page to setup alerts, change device name, and update sensor settings.

RPM
The RPM tile shows the current reading. Click on the tile to see the associated history and graph.

Machine State
The machine state tile show the state of Wind Machine.

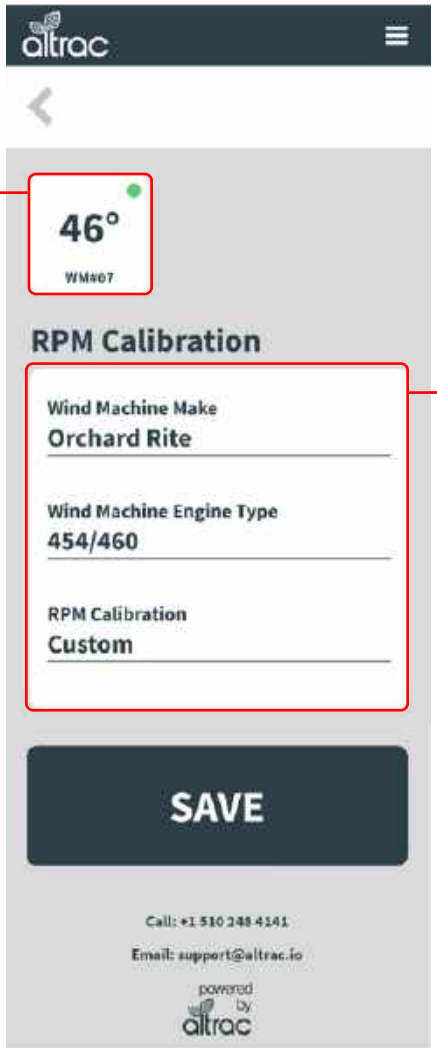
Fuel Level
The fuel level tile shows the current reading. Click on the tile to see the associated history graph.

Start and Stop Temperature
The wind machine turns ON when ambient air temperature reaches Start Temperature and OFF when the ambient temperature reaches Stop Temperature.

RPM Calibration

To calibrate the RPM reading, log into your account and click on the 'Settings' (gear icon) on the individual wind machine page. To set up an account, please contact the distributor who installed your device or support@altrac.io.

Wind Machine Tile
The tile shows which Wind Machine you are editing as well as the current status.

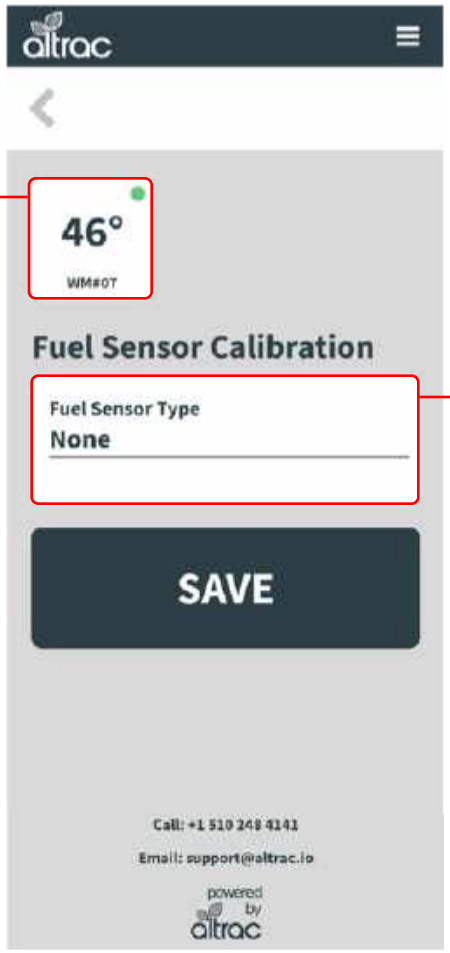


RPM Calibration
Select the wind machine make and engine type. Once chosen, the RPM will be scaled appropriately.

Fuel Sensor Calibration

To calibrate the fuel sensor reading, log into your account and click on the 'Settings' (gear icon) on the individual wind machine page. To set up an account, please contact the distributor who installed your device or support@altrac.io.

Wind Machine Tile
The tile shows which Wind Machine you are editing as well as the current status.



Fuel Sensor Calibration
Select a fuel sensor type to display the correct fuel sensor reading for your wind machine.

Installation Testing and Finaling

After installing the ST100 and wiring harness, you must test run the wind machine to ensure a successful installation.

1. Check cellular signal by navigating to the 'Settings' menu (gear icon). If the cellular signal is above 100 RSSI or you see a warning, you may need a directional antenna.

2. Attempt to start the wind machine by adjusting your 'Start Temperature' and 'Stop Temperature' in the webapp.

3. Once the wind machine is running, you should wait for several RPM readings to verify RPM is calibrated correctly.

4. If all readings are present and accurate the installation is correct.



Device and Tile States

The Altrac station has the ability to check its own operation continuously. It will notify you of the state of the machine and if a fault occurs, an error code will display in the webapp. Please identify the code displayed when inquiring about service.



Altrac strives to make issues fixable by anyone. However, some of the checks below may need to be completed by a qualitified service technician. Call a service technician if you have any doubt or servation about performing the remedy yourself.

Overview Tile

Tile	State	Description	Troubleshooting Steps
	OFF	Equipment is OFF. Physical switch is in the manual position. The machine cannot start remotely.	Move physical switch to the 'Auto' position.
	OFF - Disarmed	Equipment is OFF. Physical switch is in Auto Position but the machine is not armed to run via the Altrac app.	
	OFF- Armed	Equipment is OFF. Physical switch in Auto Position and machine is armed to run. A green dot on the tile means that the equipment will run automatically according to stored settings.	
	FULL RUN	Equipment is FULL RUN. When the overview tile is green it means that the equipment is running normally and without an issue. A green dot on the tile means that the equipment will run automatically according to stored settings. A yellow dot on the tile warns of low risk issues.	Click on the tile to view the warning.

Tile	State	Description	Troubleshooting Steps
	WARM UP or COOL DOWN	Equipment is warming up or coolings down. When the overview tile is blue it means that the equipment is starting but has not reached full run. A green dot on the tile means that the equipment will run automatically according to stored settings. A yellow dot on the tile warns of low risk issues.	Click on the tile to view the warning.
	Altrac Cellular Station Disconnected	The cellular modem is disconnected from the cloud.	1. Check that the device has power. 2. Check last cellular strength of last connection. 3. Check wiring on equipment Refer to altrac.zendesk.com ST100 troubleshooting guide for more information.
	Temperature Probe Disconnected	The temperature probe on the Altrac device is disconnected. The machine cannot run automatically since operation is triggered by temperature.	1. Check that a temperature probe is plugged in. 2. Check that the temperature probe is not damaged. Refer to altrac.zendesk.com ST100 troubleshooting guide for more information.

Controller State

Tile	State	Description	Troubleshooting Steps
	Black Tile: MANUAL/AUTO	The physical switch on the wind machine is in the 'Manual' position. This machine will not run based on start and stop temperatures and cannot be started remotely.	If the switch is in the AUTO position: 1. Check Altrac wiring including ground wire. 2. Check switch is functional. 3. Confirm firmware is installed correctly.
	White Tile: MANUAL	The physical switch on the wind machine is in the 'Auto' position. This machine can be started and stopped remotely, but will not run automatically based on start and stop temperatures.	
	White Tile: AUTO	The physical switch on the wind machine is in the 'Auto' position. This machine can be started and stopped remotely and will run automatically based on start and stop temperatures.	

Machine State

Tile	State	Description	Troubleshooting Steps
	OFF	The wind machine engine is not running.	If the wind machine engine is running but no RPM appears: 1. Check wiring from RPM sensor to Altrac device. 2. Make sure the correct firmware is installed on your Altrac device.
	FULL RUN	The wind machine engine is running at full RPM.	If the Altrac app shows FULL RUN but the engine is actually running at a lower speed, check your RPM calibration in the Altrac app device settings.
	WARM UP or COOL DOWN	The wind machine engine is running at low RPM.	If the Altrac app shows WARM UP or COOL DOWN but but the engine is actually running at full speed, check your RPM calibration in the Altrac app device settings.

RPM

Tile	State	Description	Troubleshooting Steps
	Zero RPM	Wind machine engine is not running.	If the wind machine engine is running but no RPM appears: 1. Check wiring from RPM sensor to Altrac device. 2. Make sure the correct firmware is installed on your Altrac device.
	RPM Reading	Wind machine engine is running.	If the wind machine show an inaccurate RPM reading you an adjust the calibration in the Altrac app settings for the device in question.

Air Temperature

Tile	State	Description	Troubleshooting Steps
	Temperature Shown	The temperature probe measures air temperature in the fruit zone of your crop.	
	Temperature Probe Disconnected	The temperature probe on the Altrac device is disconnected. The machine cannot run automatically since operation is triggered by temperature.	1. Check that a temperature probe is plugged in. 2. Check that the temperature probe is not damaged. Refer to altrac.zendesk.com ST100 troubleshooting guide for more information.

Battery Voltage

Tile	State	Description	Troubleshooting Steps
	Zero Voltage	Battery is not installed on the wind machine.	If battery is installed but no voltage is shown: 1. Check battery cables 2. Check fuses 3. Check Altrac wiring harness
	Voltage Shown	Normally, the Altrac device reads the battery voltage of the wind machine battery.	
	Low Voltage		1. Charge your wind machine engine battery 2. Check battery voltage with multimeter 3. Check battery cables 4. Check Altrac wiring harness

Fuel Level

Tile	State	Description	Troubleshooting Steps
	Probe Not Connected	The fuel sensor is not connected to the Altrac device.	1. Check sensor connection and wiring. 2. Ensure wind machine battery is installed and device is receiving 12VDC power. 3. Check firmware version is compatible.
	Probe Connected	The fuel sensor is connected and reading correctly.	
	Probe Overvoltage	If the fuel sensor is connect but giving a value outside of it's range, you will see the following error tile.	1. Check sensor connection and wiring. 2. Check sensor is properly connected to dial.

Banners

Tile	State	Description	Troubleshooting Steps
	Altrac Cellular Station Disconnected	Altrac Cellular Station not connected to the cloud.	1. Check that the device has power. 2. Check last cellular strength of last connection. 3. Check wiring on equipment Refer to altrac.zendesk.com ST100 troubleshooting guide for more information.
	Low Battery	Wind Machine Engine Battery is Low	1. Charge your wind machine engine battery 2. Check battery voltage with multimeter 3. Check battery cables 4. Check Altrac wiring harness
	Black Controller State Tile: MAN/AUTO	The physical switch on the wind machine is in the 'Manual' position. This machine will not run based on start and stop temperatures and cannot be started remotely.	If the switch is in the AUTO position: 1. Check Altrac wiring including ground wire. 2. Check switch is functional. 3. Confirm firmware is installed correctly.

Connectivity

The Altrac station includes a cellular modem with an embedded SIM card. Altrac negotiates with cellular networks to achieve a low price and bundle data into the cost of the device.

Altrac equipment in the United States uses LTE CATM1 (5G) networks. 2G/3G options are available upon request for other countries. The Altrac modem will automatically pick the strongest signal for its connection.

Update Frequency
ST100 OFF: 15 minutes.
ST100 ON: 2 minutes or less.
ST100 Setting Changes from Webapp: Up to 2 minutes while machine is running. Up to 15 minutes while machine is OFF.

If you need to update your ST100 immediately, you should unplug and plug back in the 12-pin Deutesch connector between the control line and the engine wiring harness. Once you plug the connector back in and the ST100 receives 12VDC power, it will reset the modem and update.

Zone 1 Networks

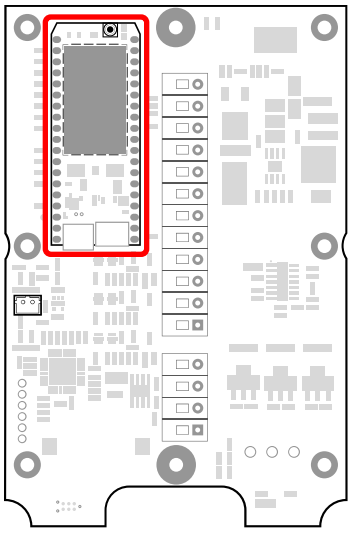
Country	Model	Carriers	2G Sunset	3G Sunset	5G Sunset
United States	ST100_3G	AT&T	2016-12-31	2021-12-31	
	ST100_3G	T-Mobile (USA)	2020-12-31		
	ST100_5G	AT&T			
Canada	ST100_3G	Rogers AT&T Wireless	2020-12-31	2025-12-31	
	ST100_5G	Rogers Wireless			
	ST100_5G	Telus			
	ST100_5G	Bell Mobility			
Mexico	ST100_3G	Telefonica Moviles Mexico	2019-12-31		
	ST100_5G	AT&T			
Australia	ST100_3G	Telstra MobileNet	2016-12-31	2024-06-30	

Zone 3 Networks

Country	Carriers	Model	2G Sunset	3G Sunset
United Kingdom	Telefonica O2 UK	ST100_3G		
Afghanistan	Roshan (TDAC)	ST100_3G		
Algeria	Mobilis	ST100_3G		
Argentina	Movistar Argentina	ST100_3G		
Aruba	Setar	ST100_3G		
Austria	T-Mobile Austria	ST100_3G		
Azerbaijan	Azercell	ST100_3G		
Bahrain	Batelco	ST100_3G	No Plan	No Plan
Bangladesh	GrameenPhone	ST100_3G		
Belarus	Velcom (Mobilkom)	ST100_3G		
Belgium	Base (KPN)	ST100_3G		
Bolivia	NuevaTel (Viva)	ST100_3G		
Bosnia and Herzegovina	BH Telecom	ST100_3G		
Botswana	Mascom Wireless	ST100_3G		
Brazil	Vivo	ST100_3G		
Bulgaria	Vivacom (Vivatel)	ST100_3G		
Cabo Verde	T+ Telecomunicações	ST100_3G		
Cameroon	MTN Cameroon	ST100_3G		
Chile	Movistar Chile	ST100_3G		
China	China Mobile	ST100_3G	No Plan	2020-01-01
Colombia	Colombia Telefonica	ST100_3G		
Congo (Brazzaville)	MTN Congo	ST100_3G		
Costa Rica	Telefonica Costa Rica	ST100_3G		
Côte d'Ivoire	MTN Cote d'Ivoire	ST100_3G		
Croatia	VipNet (Mobilkom)	ST100_3G		
Cyprus	Cytamobile-Vodafone	ST100_3G		
Czechia	Telefonica (O2) Czech Republic	ST100_3G		
Denmark	Telenor AS	ST100_3G		2021-12-31
Ecuador	Otecel S.A. Ecuador	ST100_3G		
Egypt	MobiNil	ST100_3G		
El Salvador	Telefonica Moviles El Salvador	ST100_3G		
Estonia	TeleTwo Estonia	ST100_3G		
eSwatini	MTN Swaziland	ST100_3G		
Finland	TeliaSonera Finland	ST100_3G		
France	Bouygues Telecom France	ST100_3G		
Gabon	Airtel (Zain) Gabon	ST100_3G		
Georgia	Geocell Georgia	ST100_3G		
Germany	Telefonica O2 Germany	ST100_3G		
	Telekom Deutschland GmbH	ST100_3G	2020-12-31	2020-12-31
Ghana	Vodafone Ghana (ONEtouch)	ST100_3G		
Greece	Wind Hellas	ST100_3G		
Greenland	Tele Greenland	ST100_3G		
Guatemala	Telefonica Moviles Guatemala	ST100_3G		
Guinea	GNBSB-Spacetel MTN Guinea Bissau	ST100_3G		
Guyana	Digicel Guyana	ST100_3G		
Hong Kong	CSL Ltd	ST100_3G		
	HKT	ST100_3G		
Hungary	Telenor Hungary	ST100_3G		2020?
Iceland	Siminn (On-Waves)	ST100_3G		
Indonesia	Telkomsel Indonesia	ST100_3G		
Ireland	H3G Ireland	ST100_3G		
	O2 Ireland	ST100_3G		
Israel	Partner Communication Israel	ST100_3G		
Italy	Wind Italy	ST100_3G		
	Hutchison 3G Italy	ST100_3G		
Jamaica	Digicel Caribbean	ST100_3G		
Japan	NTT DoCoMo	ST100_3G		Late 2020s
Jersey	Jersey Telecoms UK	ST100_3G		
Kazakhstan	K-Cell Kazakhstan	ST100_3G		
Kenya	Celtel Kenia Ltd	ST100_3G		
Kuwait	Watanya Telecom Kuwait	ST100_3G		

Country	Carriers	Model	2G Sunset	3G Sunset
Latvia	TELE2 Latvia	ST100_3G		
Liechtenstein	Mobilkom Liechtenstein	ST100_3G		
Lithuania	TELE2, Lithuania	ST100_3G		
Luxembourg	TANGO Mobile SA	ST100_3G		
Macao	CTM Macau	ST100_3G	2015-06-30	
Macedonia	T-Mobile Macedonia	ST100_3G		
Madagascar	Celtel Madagascar	ST100_3G		
Malaysia	Digi Telecom Malaysia	ST100_3G		
Malta	Go Mobile Malta	ST100_3G		
Moldova, Republ	Orange Moldova	ST100_3G		
Mongolia	Unitel Mongolia	ST100_3G		
Montenegro	Telenor Montenegro	ST100_3G		2020?
Montserrat	Cable & Wireless Montserrat	ST100_3G		
Morocco	Medi Telecom	ST100_3G		
Mozambique	Vodacom Mozambique	ST100_3G		
Netherlands	KPN B.V.	ST100_3G		2022-01-01
Netherlands Antil	Digicel Curacao	ST100_3G		
New Zealand	Spark New Zealand	ST100_3G	2012-07-31	No Plan
Nicaragua	Telefonia Celular de Nicaragua	ST100_3G		
Niger	Celtel Niger S.A.	ST100_3G		
Nigeria	Celtel Nigeria	ST100_3G		
Norway	Telia Sonera Norge	ST100_3G		2019-01-01
Oman	Omantel	ST100_3G		
Pakistan	Mobilink Pakistan	ST100_3G		
Panama	Telefonica Moviles Panama S.A.	ST100_3G		
Peru	Telefonica Moviles Peru	ST100_3G		
Philippines	Globe Telecom Philippines	ST100_3G		
Poland	Polkomtel Poland	ST100_3G		
Portugal	TMN Portugal	ST100_3G		
Qatar	Q-TEL Qartar	ST100_3G		
Réunion	Reunión-Mayotte-SRR-MTM-Societe Reunionnaise du Radio	ST100_3G		
Romania	Orange Romania SA	ST100_3G		
Russia	Vimpelcom (Beeline)	ST100_3G		
Rwanda	MTN RwandaCell	ST100_3G		
Saint Kitts and N	Cable & Wireless St Kitts & Nevis	ST100_3G		
Saudi Arabia	Al Jawal	ST100_3G		
Serbia	Telenor Serbia	ST100_3G		2020?
Seychelles	Airtel (Seychelles)	ST100_3G		
Singapore	StarHub Mobile Pte Ltd.	ST100_3G		
Slovakia	Telefonica O2 Slovakia	ST100_3G		
Slovenia	Si.Mobil Slovenja	ST100_3G		
South Africa	Cell C (PTY) Ltd.	ST100_3G		
South Korea	SK Telecom	ST100_3G		
Spain	Telefonica Moviles España	ST100_3G		
Sri Lanka	Dialog Telekom Ltd	ST100_3G		
Suriname	Digicel Suriname	ST100_3G		
Sweden	Telenor Sverige AB	ST100_3G		2020-12-31
Switzerland	Sunrise	ST100_3G	2018-12-31	2024-12-31
Taiwan	Taiwan Mobile Co., Ltd.	ST100_3G	2017-06-30	2017-01-07
Tanzania	Celtel Tanzania Ltd.	ST100_3G		
Thailand	Total Access Communic.Thailand	ST100_3G		
Trinidad and Tob	Telecom.Trinidad and Tobago Lim.	ST100_3G		
Tunisia	Orange Tunisie	ST100_3G		
Turkey	AVEA Iletisim Hizmetleri A.S.	ST100_3G		
Uganda	MTN Uganda	ST100_3G		
Ukraine	Kyivstar Ukraine	ST100_3G		
United Arab Emir	du - Emirates I.T.C.	ST100_3G		
United Kingdom	Telefonica O2 UK	ST100_3G		
Uruguay	Movistar Uruguay	ST100_3G		
Venezuela	Telefonica Moviles Venezuela	ST100_3G		
Vietnam	Viettel Mobile	ST100_3G		

Cellular Modem



Each ST100 includes a cellular modem. The modem automatically connects to the cellular network when the device has power. You may see the following states below.

The cellular modem is located on the top right of the circuit board. The modem is removeable in case of damage or obsolesence. To remove, detatch the antenna by gently pulling up on the gold u.Fl antenna connector. Without exterting pressure on the top of the modem, pull up on the sides to remove the modem from its headers.

Standard Modes

The states below are the normal sequence you will see on startup.



Looking for Internet
If your device is blinking green, it is trying to connect to the cellular tower.



Connecting to the Cloud
When the device is in the process of connecting to the cloud, it will rapidly blink cyan. You often see this mode when you first connect your device to a network, after it has just blinked green.



Connected
When it is breathing cyan, your device has successfully connected to the Internet.

Troubleshooting Modes



Cloud Not Connected
When your device is connected to cellular but not to the cloud, the LED will breath green.

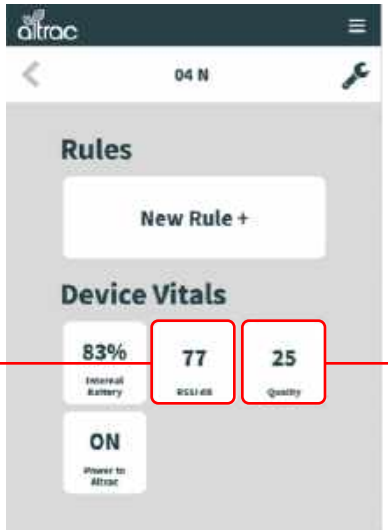
If you see the above state or any other color, please contact support@altrac.io

The modem is the most sensitive part of the ST100. Removing the antenna connector or modem without the proper tools could void your warranty. Contact Altrac. We're happy to help.

Internal Antenna

The ST100 has an antenna mounted inside the enclosure. Because of this, mounting location of the ST100 can affect cellular signal strength. You can check the cellular signal health of your device by navigating to the settings tab in your Altrac account.

Signal Strength
Cellular signal strength of the device. Lower numbers are better.



Quality
Cellular signal quality. Higher numbers are better.

Signal Strength
RSSI stands for 'Relative Signal Strength Indicator'. Any reading under 100 is acceptable and should not have trouble connecting. Anything over about 105 RSSI will be unstable with weather / congestion variances.

Signal Quality
Typically a quality of 0 means noise levels are high enough to not have a connection. A quality value of 49 is a solid noise free connection. In some scenarios a device will have a good RSSI reading, and bad signal quality reading; in this case you may need an external antenna (CN102).

Mounting Location
- As high above the ground as practical.
- In an area clear of vegetation.
- Orient the right or left side of the ST140 to face the cellular tower.

Troubleshooting Cellular Signal

- Rotate your device 90 degrees (keep connectors facing the ground). Often, the internal antenna picks up signal better in certain orientations.
- Metal weakens cellular signals. If possible keep the device away from metal, or if signal is weak, change location of the device to the opposite side of a metal cabinet.
- Add an external antenna to the device.

External Antenna

If you have followed the troubleshooting tips on the previous page and still have weak signal, you likely need an external antenna to boost the cellular signal strength. The antenna comes with two components: the antenna and the connector. Contact your distributor or sales@altrac.io to purchase the antenna.



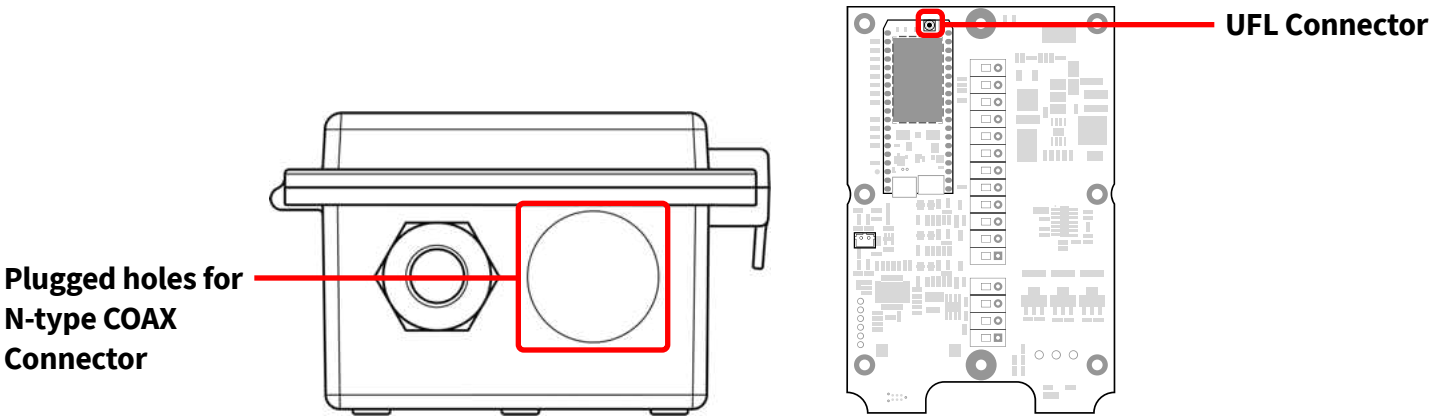
External Antenna



N-Type to UFL Connector

Installation

You will need to install the N-type connector to the bottom of the ST100.
Step 1: Remove the plug from one of the pre-drilled holes on the bottom of the ST100 enclosure.
Step 2: Use the nut to attach the N-type connector (large silver side) to the enclosure.
Step 3: Remove the existing internal antenna UFL connector (smaller and gold side).
Step 4: Attach the new UFL connector to the cellular modem.



Mounting

- As high above the ground as practical.
- Above any metal sheeting or cabinet.
- In an area clear of vegetation.

Backup Battery

The ST100 includes a backup battery designed to power the device for up to 3 hours during a power outtage. This battery will not allow the wind machine to continue operating, but it does allow the ST100 to notify the user when their Wind Machine has lost power.

Specifications:
Altrac Part Number: PW100
Battery Type: Lipo
Continuous Discharge Rating: >= 3A
Nominal Voltage: 3.7v
Nominal Capacity: >= 1800mAh
Rechargeable: Yes



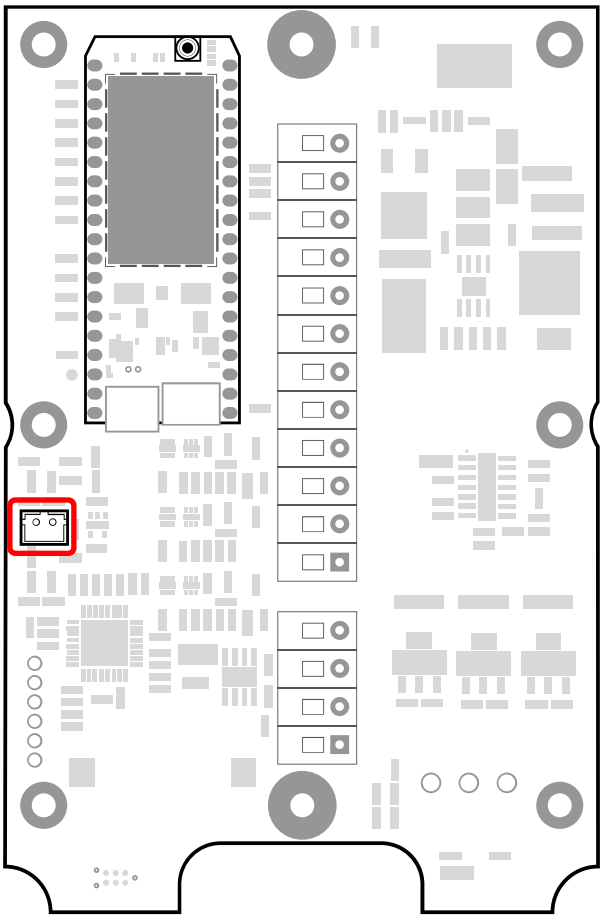
Protection Circuit: Mandatory. If you do not use a battery with an integrated PCB protection circuit, the warranty on the ST100 is void.

Replacement:
Altrac offers replacements batteries.
Contact sales@altrac.io for more information.

Installation:
1. Remove the existing battery by prying the battery loose from the side of the enclosure.

2. Attach the new battery by removing the VHB tape backing and pressing firmly on the side of the enclosure.

3. Reattach the small white connector to the circuit board highlighted in red on the line drawing to the right.



Warranty

One Year Limited Warranty

1. What is covered by this warranty
All Altrac hardware comes with a 1-year warranty from date of purchase. During the warranty period, Altrac will replace or repair, at no charge, any product or parts of product that prove defective due to improper materials or workmanship under normal use. Altrac must be notified in writing of any product defect.

2. What is not covered by this warranty
Altrac does not warrant (a) defects caused by failure to provide a suitable installation environment for the product; (b) damage caused by use of the product for purposes other than those for which it was designed; (c) damage caused by disasters such as fire, flood, and lightning; (d) any other abuse or misuse by the purchaser.

3. Disclaimer of warranty
The foregoing warranties are in lieu of all other warranties, express or implied, including but not limited to the implied warranties of merchantability and fitness for a particular purpose.

4. Limitation of remedies
In no event will Altrac be liable for any special, incidental, or consequential damages based on breach of warranty, breach of contract, negligence, strict liability, tort, or any other legal theory. Damages that Altrac will not be responsible for include, but are not limited to: loss of profits, loss of revenue, loss of use of the product or any associated equipment, loss or damage to crops, cost of any substitute equipment, facilities, or services, downtime, the claims of third parties including customer, and injury to property.

5. No other warranties
This statement is intended and understood to be the complete and exclusive warranty with respect to the product that is the subject of this sale and said warranty supersedes all oral or written prior agreements and all other communications between the parties relating to the subject matter of this warranty, including statements made by salespersons. No employee of Altrac or any other party is authorized to make any warranty in addition to those set forth herein.

6. Mandatory Arbitration
All disputes arising under this warranty, including those which may otherwise have been determined in a court of law, shall be submitted to and determined solely through binding arbitration as determined by Altrac.